

Group of 20 (G20) Study Guide

**AGENDA: International Economic Cooperation:
Assessing the impact of regional
instability on global energy security, supply chain
resilience and macro-economic stability.**



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1. Letter From the Executive Board

It is our distinct pleasure to invite all of you to this simulation of the Group of 20 (G20), at this edition of Calcutta Boys' School Assembly of Nations, 2026. Please note that this background guide is for your own perusal, to provide direction to your research. The real efficacy of this, however, lies in the use of this guide as a reference; evaluate the potential problem statements; and propose practical solutions that can be covered under the spectrum of this agenda, while incorporating them within your deliberations.

Understanding the complexity, as well as the significance of this agenda, the Executive Board strongly recommends all of you to come prepared and well researched in committee, and at the same time, we expect you to participate throughout the course of the committee sessions. We look forward to putting you all in various diplomatic practices that might be deemed necessary to provide more substantiation to the deliberations at hand. We sincerely believe that all of you will come prepared, well researched, and motivated enough to discuss the agenda, and its nuances and brainstorm together to explore solutions.

Our goal for you in this committee is to provide you with an enriching experience by helping you to navigate through the art of diplomacy, and at the same time, see you deliberate on real life problems, and global challenges that currently exist in the real world. We also expect all delegates to maintain courteous behavior throughout the course of the entire conference, while looking at real life situations from a diplomatic standpoint. Looking forward to meeting all of you at Calcutta Boys' School Assembly of Nations, 2026.

For any queries/clarifications/enquires, email at g20.cbsaon@gmail.com

With regards,

Mallar Mitra
(Chairperson)

Kinshuk Trivedi
(Co-Vice Chairperson)

Agneya Goswami
(Co-Vice Chairperson)

2. Group of 20 (G20)

Founded in 1999, the 'Group of 20' (G20) is the latest of post-WWII initiatives aimed at international coordination of economic policy. It is an informal intergovernmental forum that was established to increase dialogue on global economic growth and prosperity issues after the Asian financial crisis. Following the meeting of the Group of Seven's (G7) Ministers of Finance and Governors of Central Banks, the G20 was initially launched to focus on the international financial instability that had found its roots in Asia in 1997. The G7 has been criticized for its lack of legitimacy to make decisions that could affect the whole world because of limited membership and narrow representation. Together with this criticism and after the international community's disappointment about how the G7 handled global economy problems, the G20 it was supposed to include middle-income countries and those with systemic economic influence in finding solutions to economic problems affecting everyone.

The decisive transformation came in 2008. Confronted with the most severe financial crisis since the Great Depression, the United States convened the first G-20 Leaders' Summit in Washington in November 2008, elevating the forum from a technocratic gathering of finance officials to a summit of heads of state and government. The 2008 to 2009 episode remains the G-20's defining achievement and its most instructive precedent. Faced with a synchronised collapse in demand and a seizure of global credit markets, the leaders coordinated fiscal stimulus, resisted the competitive protectionism that had deepened the Depression of the 1930s, tripled the lending capacity of the International Monetary Fund, and established the Financial Stability Board to monitor systemic risk. The coordination held because the threat was common, immediate, and unambiguous, and because failure to cooperate carried a visible and shared cost. The agenda before this committee will require delegates to assess whether those conditions can be reconstructed for a slower-moving and more

diffuse set of threats, or whether the 2008 model was a product of circumstances that do not generalise.

The G20 is composed of most of the world's largest economies, including both industrialized and developing countries. Specifically, the G20 comprises of 19 sovereign countries (Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, Mexico, Republic of Korea, Republic of South Africa, Russia, Saudi Arabia, Turkey, United Kingdom and the United States of America), the European Union (EU) and the African Union (AU). Although Spain is not an official member of the G20, it is considered a permanent guest to the summits. The Presidency also invites guest countries and several international and regional organizations, to grant the forum an even broader representation. Each year, the G20 Summit is chaired by a different member country, selected from a different regional grouping of countries. Known as the Troika, the chair of a summit

Structure, Composition and Mandate of the G20.

The G20 mandate, however, has shifted after its upgrade to the level of Heads of State/Government from focusing on preventing and resolving financial crises to longer-term economic issues. Following the financial crisis of 2007-2008 and subsequent improvements of world economic conditions in 2009, the G20 objectives were formulated more explicitly. As of the 2009 G20 Summit in Pittsburgh (U.S.), the objectives are now to create strong, sustainable and balanced economic growth. To reach this goal, the 2011 G20 Summit in Cannes agreed that the G20 has the responsibility to “coordinate their policies and produce political agreements that are very important in addressing challenges due to conditions of global economic interdependence”.

The main focus of the G20 is macroeconomic policy, but the topics discussed may change based on

the needs and concerns of member countries. Therefore, although economic and financial coordination remain the focus of the summits, fair and sustainable development, the future of work, digital economy, terrorism and global health have been recurring topics as well.

The G-20 has no permanent secretariat, no headquarters, and no standing budget, an institutional design choice that distinguishes it sharply from treaty-based bodies. Continuity is instead provided by the troika, a rolling grouping of the immediate past, current, and immediately succeeding presidencies, which collaborates to maintain agenda coherence across annual cycles. The presidency rotates annually and carries substantial agenda-setting power, since the host determines thematic priorities, convenes the working groups, and drafts the leaders' declaration. This rotation has recently become a site of visible strain. The 2025 presidency was held by South Africa, whose theme of "Solidarity, Equality, Sustainability" foregrounded debt relief, climate finance, and just energy transitions, and which hosted the first G-20 summit on African soil in Johannesburg in November 2025. That summit was marked by the absence of several leaders and a boycott by the United States, which objected both to the thematic framing and to what it characterised as agenda overload. The 2026 presidency passed to the United States, which has reoriented the forum toward what it calls its core economic mission, naming deregulation, affordable and secure energy supply chains, and technological innovation as its priorities, and scheduling the leaders' summit in Miami. Delegates should understand that the troika mechanism, designed to guarantee continuity, is now itself a contested object, and that the question of who sets the agenda is not separable from the substantive agenda this committee will debate.

The G-20 operates through two parallel workstreams. The Finance Track convenes finance ministers and central bank governors and addresses macroeconomic coordination, financial regulation, international financial architecture, taxation, and debt. The Sherpa Track, named for the personal representatives whom leaders appoint to negotiate on their behalf, addresses the broader non-financial agenda, including development, energy, trade, climate, health, and digital governance. Beneath these tracks sit a proliferating structure of working groups and task forces, which by the 2025 presidency numbered more than twenty working groups alongside multiple task forces, and a set of engagement groups that channel input from business, civil society, labour, think tanks, and other constituencies. For the purposes of this agenda, delegates should recognise that energy security and supply chains straddle both tracks, since they are simultaneously questions of macroeconomic stability, which is Finance Track terrain, and questions of physical and industrial policy, which is Sherpa Track terrain. This cross-cutting character is itself a source of the coordination difficulty the committee must confront.

3.

3. What an Economic Forum Means by "Regional Instability"?

For the G-20, instability matters not as a moral or security category but as a source of disruption to the flows on which the integrated global economy depends: flows of energy, of goods and components, of capital, and of information. A civil war that disrupts none of these flows is, in the cold terms of this mandate, economically peripheral; a localized confrontation that closes a strait through which a fifth of the world's seaborne oil passes is economically central, even if it produces few casualties. Delegates should resist the instinct to rank instabilities by their humanitarian severity and instead rank them by their economic transmission: the size of the flow disrupted, the availability of substitutes, the speed at which the disruption propagates, and the buffers available to absorb it.

Recognise that energy security, supply-chain resilience, and macroeconomic stability are not three separate problems but three readings of a single underlying condition. That condition is the cost a system incurs when its reliance on geographically concentrated dependence is exposed to instability.

A globally integrated economy sources energy, components, and finance from the lowest-cost locations available. This is enormously efficient, and during periods of stability it appears costless. But efficiency of this kind is purchased by giving up the ability to substitute cheaply: the system has, in effect, sold the option to switch suppliers or routes quickly in exchange for lower steady-state prices. Regional instability does not necessarily change the physical configuration of the system, at least not at first. What it changes is the probability that a given concentrated dependence will be disrupted, and in doing so it forces a revaluation of every similar dependence across the system. This is why a shock confined to a single corridor or a single supplier can move global aggregates far out of proportion to the physical volume actually interrupted: the disruption does not merely halt one flow, it informs every actor that a

whole class of comparable flows is more fragile than they had priced.

From this vantage, three terms are worth carrying through the debate. The first is the chokepoint, a point of physical concentration, a strait, a canal, a pipeline, a single dominant supplier or fabricator, through which a disproportionate share of a critical flow must pass and for which substitutes are scarce or slow. The second is resilience, properly understood as the deliberate and costly repurchase of the substitution options that efficiency sold: redundant suppliers, strategic reserves, diversified routes, held inventory. Resilience is not free, and one of the committee's central tasks is to decide who pays for it. The third, which Section 10 develops, is the cooperation paradox: the observation that instability simultaneously raises the value of cooperation and erodes the conditions that sustain it. With these terms established, the three pillars can be examined in turn.

4. *PILLAR I: GLOBAL ENERGY SECURITY*

Energy security is conventionally defined as the uninterrupted availability of energy sources at an affordable price, a formulation that usefully separates two distinct dimensions that are often conflated. The first is security of supply, the physical question of whether energy can be obtained at all. The second is security of price, the economic question of whether it can be obtained at a cost that does not itself destabilise the economy. A regional instability can leave physical supply technically intact while inflicting severe damage through price, as occurred in Europe in 2022, and delegates should be precise about which dimension a given proposal address, because the policy instruments differ. Strategic reserves address physical availability; demand management and price caps address affordability; diversification addresses both but slowly and at cost.

Energy markets are uniquely exposed to regional instability because energy is physically concentrated in

its production and constrained in its transit. A small number of maritime chokepoints carry an outsized share of global energy trade. The Strait of Hormuz, between the Persian Gulf and the Gulf of Oman, carries on the order of a fifth of the world's petroleum liquids and a comparable share of liquefied natural gas, with no fully adequate bypass for the volumes involved. The Strait of Malacca channels the bulk of crude oil bound for East Asia. The Bab-el-Mandeb strait and the Suez Canal together constitute the principal corridor between Gulf and Mediterranean and European markets. The Turkish Straits and the Danish Straits carry significant volumes from the Black Sea and Baltic respectively. Each of these is a point at which a regional security event can convert directly into a global price shock, and the concentration is structural rather than accidental, since the geography of hydrocarbon deposits and the economics of maritime transport jointly dictate it.

Supplier concentration compounds corridor concentration. A handful of states, principally in the Gulf, Russia, and the United States, dominate marginal oil and gas supply, and the willingness of any of them to withhold supply, whether for political leverage or as a consequence of internal instability, moves the global price. The committee should treat both forms of concentration as instances of the same vulnerability: a flow on which many depend passes through a node controlled by few.

Any evaluation of the G4 contribution to energy governance should begin with a discussion of what 'energy governance' really means. Governing energy is often equated with ensuring 'energy security,' traditionally defined as having adequate and reliable supplies of energy (read: oil and gas) at reasonable prices. This is, for example, the definition employed by the Paris-based International Energy Agency (IEA), an offshoot of the Organization of Economic Cooperation and Development (OECD). Thus conceived, energy security revolves around two key concepts: availability and affordability.

Yet, this is actually a very narrow approach of energy security. For starters, it does not take into account the interests of the main oil- and gas-exporting countries. While consumers are interested in the security of supply, producers worry more about security of demand and, especially, revenues. When energy security is on the table of the G20 leaders, the representatives of Russia and Saudi Arabia will strive for demand security to be taken into account. This seems like a fair point because these countries need to plan their investment policy in the long term. Moreover, on the face of it, security of supply and demand are just two sides of the same coin.

In reality, there are key distributional issues that pit hydrocarbon exporters and importers off against each other. One is the ‘rent contest’ between the governments of oil and gas exporters and importers. It is often overlooked that the G7 governments earn more from oil taxes than the governments of the Organization of Petroleum-Exporting Countries (OPEC) earn from oil sales. Over the period 2009-13, for example, OECD countries earned an average of \$1,082 billion per year from oil taxes, whereas OPEC members earned an average of \$966 billion per year (OPEC, 2014). This is an issue that has long been, and continues to be, a thorn in the eye of the oil cartel.

Another, even more divisive issue are the ambitious decarbonization policies that (some) G7 countries are announcing and implementing. In the long run, these policies could translate into shrinking energy export revenues for major hydrocarbon producers. Western consumer countries have often taken a hypocritical stance by announcing large decarbonization schemes, on the one hand, while asking OPEC to pump more oil when oil prices are high, on the other. Some would add the proposed policies of ‘energy independence’ in the United States (US) to this list of divisive issues but energy pundits agree

that this is essentially an empty slogan and it does not really upset major energy exporters. The recent rhetoric in the EU to ‘get off Russian gas’ , not for reasons of environmental zeal but for geostrategic concerns, is a far more credible threat.

One way to bridge this divide between energy importers and exporters is through the concept of ‘energy services.’ Households and (most) companies do not need energy sources (say, a barrel of oil, a ton of coal, or a cargo of LNG). Instead, they require the services that these sources can yield: cooking, lighting, refrigeration, heating, mobility and others. Putting energy services central in the discourse on energy governance is one way in which the producer consumer schism can be overcome. The United Nations (UN) have started to make serious headways on this front with their Sustainable Energy for All (SE4ALL) initiative, launched by the UN General Assembly in late 2010. This process has led to the inclusion of energy into the Sustainable Development Goals, more precisely as Goal 7 to ensure access to affordable, reliable, sustainable and modern energy for all.

The energy governance agenda does not end there, however. Besides the availability and affordability of energy services (and not energy sources, as just explained), policy-makers would do well bring in the sustainable development dimension. This draws attention to the issues of ‘accessibility’ and ‘acceptability’. On energy access, world leaders urgently need to address the problem of electricity deprivation. It is simply unacceptable that 1.3 billion people – that is, a population five times the size of the US – still lacks access to electricity in their homes, and thus lacks access to elementary energy services. The element of ‘acceptability’ refers to the various externalities our energy system incurs, most notably the environmental dimension. Given the fact that fossil fuels are the main culprit for CO₂ emissions, climate change should be an integral part of the energy governance agenda. While climate

change is arguably the most severe and urgent adverse environmental shock generated by our energy system, the list of energy-induced environmental problem is longer and includes issues like air pollution (caused by emissions of substances such as SO₂, NO_x and particulate matter), water quality and availability, and land use change.

In sum, there are four major goals in global energy governance, commonly referred to as the '4 A's': ensuring the availability, affordability, accessibility and acceptability of energy services. Action is sorely needed as there are critical signposts that these goals are currently not being delivered. Think of the growing anxiety about energy security around the world, oil prices hovering above \$100 a barrel on average in the period 2011-2014 (though they have markedly dropped since June 2014), persistent energy poverty in Asia and Sub-Saharan Africa, and the crossing in May 2013 of the 400 parts per million (ppm) threshold of greenhouse gas concentrations in the atmosphere (WMO, 2013). The overall energy governance agenda is thus quite broad, but it also crosses many policy silos and involves balancing across competing priorities. For example, when the members of the International Energy Agency (IEA) released strategic oil stocks in 2011 to stabilize the oil market in the wake of the Libya war (availability goal), they also pumped more carbon dioxide in the atmosphere (acceptability goal).

There is certainly no lack of international energy organizations and forums to deal with common energy challenges ranging from the IEA, OPEC, the International Energy Forum (IEF), the Energy Charter Treaty (ECT), the International Renewable Energy Agency (IRENA), the United Nations Framework Convention on Climate Change (UNFCCC), and many others.³ Yet there are two major problems with this global energy architecture. The first is: rising and developing countries are severely underrepresented. China and India, countries that are contributing the most to the world's growing oil demand, are still no members of

the IEA.⁴ The second problem is: no one is in charge. There is a lot of fragmentation and these organizations often try to steer us in different directions. For example, the IEA is concerned with energy security and tries to stabilize fossil fuel markets while the UNFCCC is basically trying to destroy these markets. These gaps represent a huge opportunity for the G4 system to take a seat in the cockpit of global energy governance.

The G20 leaders' summit has much more potential to act on the 4 A's outlined earlier. The G20 has worked on energy since it was upgraded to the apex forum of global economic governance in Pittsburgh in 2009. At that summit, the G20 leaders pledged to phase out inefficient fossil fuel subsidies because they 'encourage wasteful consumption, reduce our energy security, impede investment in clean energy sources and undermine efforts to deal with the threat of climate change' (G20, 2009). In spite of this commitment, fossil fuel subsidies still amounted to \$550 billion in 2013 according to the IEA, up from \$300 billion in 2009 and outmatching renewable energy subsidies by a ratio of four to one (IEA, 2014). A recent study from the IMF even reckoned that global fossil fuel subsidies amount to the monumental figure in 2015 of 5.3 trillion dollars. That is equivalent to 6.5 percent of global GDP. The difference with the figures of the IEA can be explained by the fact that the IMF also considers the environmental and social costs of our fossil fuel use to be a form of 'subsidy.' It would be wrong to conclude from this apparent failure that the G20 cannot act as a relevant forum in global energy policy. The group still represents over 70% of global energy consumption and emissions. It is one of the rare global forums where energy producers and consumers, rising and established powers meet on an equal footing and with a broad agenda. While the G20 has not yet delivered on its pledge to eliminate fossil fuel subsidies, one could say that this was the wrong energy policy measure to start with at the G20. It might have been better to start with softer policy measures, like developing standards and improving data transparency, issues where national interests at the G20 table do not collide. Recently, the energy work of the G20 has been galvanized through the Energy

Sustainability Working Group (ESWG), created under Russia's chairmanship of the G20 in 2013. This Working Group, which was co-chaired by India and Australia, submitted its final report to the leaders of the G20 in November 2014, on the eve of the G20 Brisbane summit. At that summit, the G20 leaders for the first time dedicated a full session to discuss global energy issues, thereby anchoring energy issues firmly onto the G20's agenda. In October 2015, also for the first time, the G20 will organize an energy ministerial meeting in Istanbul, Turkey.

The Brisbane discussions on energy yielded two noteworthy declarations. The first was the G20 Energy Efficiency Action Plan, a very detailed 10-page document covering energy efficiency in vehicles, buildings, networked devices, industrial processes and electricity generation. This plan of action foresees a large role for the International Partnership for Energy Efficiency Cooperation (IPEEC), hosted at the IEA headquarters in Paris, to oversee the commitments.

The second outcome was the G20 Principles on Energy Collaboration (G20, 2014c). Contrary to the Efficiency action plan, this document was much shorter and more abstract. The most important innovation in this document is 'a common understanding that the international energy architecture needs to reflect better the changing realities of the world energy landscape.' The ESGW commissioned two surveys which revealed that more can be done to strengthen collaboration between existing international energy organizations, that developing and emerging countries need to be better represented, and that energy access and poverty should get more attention from existing organizations. However, at their Brisbane summit, the G20 leaders failed to move beyond general statements like making international organizations 'more representative' and enhancing 'coordination between international energy institutions.'

The G20's mandate in the domain of energy security derives from its foundational role as a forum for financial and economic stability, and has been progressively elaborated through the Energy Sustainability Working Group (ESWG) established in 2009, the G20 Principles on Energy Security adopted at the 2013 Saint Petersburg Summit, and subsequent presidencies' commitments to diversification, resilience, and transition. The Strait of Hormuz crisis exposes several critical analytical fault lines in how the G20 has operationalized this mandate.

The 2022 Weaponisation of Russian Pipeline Gas

The structural tension between the long-run imperatives of the energy transition and the short-run imperatives of energy security was already present before the 2026 Hormuz crisis - Russia's 2022 invasion of Ukraine had already compelled a number of European G20 members to extend coal plant lifetimes, accelerate LNG import terminal construction, and conditionally relax carbon accounting frameworks for emergency procurement. The 2026 crisis has intensified this tension to a near-breaking point. The dominant analytical framing of the past decade - that energy security and energy transition are complementary goals, because diversification toward renewables reduces dependence on geopolitically volatile fossil fuel supply chains - is empirically correct in the long run but operationally insufficient in the short run, because the transition is measured in decades while supply disruptions are measured in weeks and months. The G20, as a body that houses both the world's largest fossil fuel producers and the world's most ambitious renewable energy transitions, must navigate this tension with an analytical clarity that neither dismisses the transition imperative nor treats it as an adequate substitute for the immediate management of a fossil fuel supply emergency.

The European gas crisis of 2022 is the clearest available demonstration of how supplier concentration, accumulated quietly over years, can be weaponised in weeks. The structural precondition was dependence built up over a decade. The Russian share of European Union gas supply rose from approximately thirty percent in 2010 to above forty-five percent by 2019, a concentration that appeared economically rational while it remained costless, since Russian pipeline gas was cheap and, until the moment it was not, reliable. Europe had, in the framework's terms, sold a very large quantity of substitution options at a very low premium.

Following Russia's invasion of Ukraine, that dependence was severed. Russian piped gas exports to OECD Europe fell by roughly fifty percent year on year in 2022, a contraction of about eighty-three billion cubic metres, dropping flows to their lowest level since the mid-1980s, according to the International Energy Agency. The price response was sharply nonlinear, which is the analytically critical point. The Dutch Title Transfer Facility benchmark, the reference price for European gas, rose steeply through 2022 to a peak in late August, and the spread between European and global liquefied natural gas benchmarks reached on the order of thirty-five euros per megawatt-hour over the summer, reflecting not a global scarcity of gas but an intra-European competition for scarce regasification and pipeline capacity. The bottleneck was physical: north-western Europe could not move alternative supply to where it was needed quickly enough, so price was forced to do the rationing that infrastructure could not.

The energy transition-security paradox can be stated precisely: the long-run solution to energy insecurity (decarbonization and renewable deployment) requires a decades-long transition during which fossil fuel dependence persists; the short-run response to fossil fuel supply disruptions (emergency diversification, reserve releases, emergency production increases) can and frequently does conflict with the investment trajectories and policy frameworks necessary to sustain the long-run transition. This paradox has been

operationalized in G20 policy discourse through what can be characterized as "all-of-the-above" drift - a gradual erosion of the clear prioritization of renewable energy deployment in favor of an increasingly open-ended commitment to maximizing aggregate energy supply by all available means, with the qualifying language around fossil fuels becoming progressively more permissive with each G20 communiqué. The 2025 World Economic Forum Global Tipping Points Report observed that competition for resources is intensifying and national security - including energy security - has displaced climate objectives as the leading driver of energy policy in a growing number of governments. Record global clean energy investment of \$2.1 trillion was recorded in 2024, but growth in clean energy funding slowed compared to previous years, and the policy environment began shifting toward extended fossil fuel extraction commitments as a security hedge.

The Strait of Hormuz, the Bab-el-Mandeb, and the Premium on Passage

Where the gas crisis illustrates supplier concentration, the Gulf and Red Sea corridors illustrate the concentration of passage. The Strait of Hormuz is the single most consequential energy chokepoint on earth, and periodic episodes of confrontation in the Gulf, including tanker seizures and threats to close the strait, have repeatedly demonstrated that even the credible threat of disruption, without any actual closure, is sufficient to add a risk premium to global oil prices and to raise insurance and freight costs for the entire region. This is the option-repricing mechanism in its purest form: the market prices not the disruption that has occurred but the disruption that has become more probable. The Bab-el-Mandeb, examined in detail under Pillar Two because its primary effect has been on goods rather than energy, is also a significant energy corridor, and the committee should note that the same physical event can transmit simultaneously through energy and supply-chain channels, which is precisely why the three pillars cannot be cleanly separated.

First, the IEA's emergency response mechanism - coordination of strategic petroleum reserve (SPR) releases among member countries - was activated in response to the 2026 crisis, as it was in response to the 2022 Ukraine invasion. However, the IEA's membership does not include China or India, which are respectively the world's largest and third-largest crude oil importers. China's Strategic Petroleum Reserve, while estimated at between 80 and 100 days of import coverage, operates under an entirely opaque disclosure and release framework, making it functionally impossible to coordinate within an IEA-led emergency response. India's SPRs, housed at three underground facilities with a combined capacity of approximately 5.33 million tonnes, represent only a fraction of the buffer necessary for a prolonged disruption. The absence of the world's two most consequential non-OECD importers from the coordinated reserve release framework is not merely a procedural gap - it is a structural deficiency that reduces the effectiveness of the global emergency response architecture by an order of magnitude.

Second, the weaponization of geography by a state party raises fundamental questions about the international legal architecture governing freedom of navigation. The UN Convention on the Law of the Sea (UNCLOS) Article 38 guarantees the right of transit passage through international straits, which is non-suspendable and applies even in times of conflict between the bordering states. Iran's effective closure of the Strait through military operations against commercial shipping therefore constitutes a violation of established customary international law. Yet the G20, as a non-legally binding forum, possesses no enforcement mechanism. The crisis thus reveals the gap between the normative architecture of international maritime law and the operational reality of coercive state behaviour - a gap that can only be addressed through collective political resolve of the kind that the G20 is structurally positioned to generate, even if not to legally mandate.

Third, the crisis reveals the inadequacy of the "all-of-the-above" energy supply doctrine - the position, increasingly prominent in Western G20 policy frameworks as of 2024–2025, that energy security is best served by maximizing supply across all sources, including fossil fuels, renewables, and nuclear. The Strait of Hormuz crisis demonstrates that the problem of energy security under conditions of regional instability is not primarily one of aggregate supply volume but of supply route architecture and geographic redundancy. Even countries with substantial domestic production, such as the United States - which as of 2025 was the world's largest oil producer - experienced significant energy price disruptions, because global oil markets are integrated and price-setting occurs at the margin. Energy security, in the geopolitical sense, is therefore not achievable through production maximization alone; it requires a systematic dismantling of the geographic concentration of critical energy transit infrastructure.

The Energy Transition as a New Security Variable: Critical Minerals

The most consequential insight about critical mineral supply chain risk is not that China produces a large share of certain minerals - though it does - but that China has deliberately constructed dominance at the processing and refining stage of the value chain, which is the stage at which raw geological material is transformed into the chemically and metallurgically defined inputs that battery, motor, and grid technology manufacturers require. This distinction between mining dominance and refining dominance is analytically critical because it determines the nature and depth of the dependency. A country that mines a significant share of a given mineral but does not control its refining faces commercial displacement risk - other producers can compete for mining market share - but not structural dependency. A country that controls the refining capacity for a given mineral controls the transformation of the global ore supply into usable industrial inputs, regardless of where the ore originates. This is the

architectural position China has constructed.

According to the IEA's Global Critical Minerals Outlook for 2025, China is the leading refiner for 19 out of 20 important strategic minerals analyzed, with an average market share of approximately 70% of global processing capacity overall. Data from 2024 shows China as the leading producer or processor for roughly 99% of gallium, 95% of magnesium, 83% of tungsten, 79% of graphite, and over 69% of all rare earths. These are not marginal market shares - they represent structural control over the transformation stage of the supply chain. China controls over 90% of refining capacity for both graphite and rare earth elements and processes around 60% of global lithium and cobalt. As China-Global South Project analysis concluded: "if the world is transitioning to clean energy, it is doing so through supply chains that Beijing commands."

The architectural significance of this position is compounded by a second dimension of Chinese dominance that receives less analytical attention: control over the production equipment and process technology for refining operations. In 2025, several Chinese equipment suppliers limited shipments abroad, making it harder for competitors in the United States and Europe to build their own refining systems. These export limits are part of a broader strategy to retain the high-value manufacturing activities within China's industrial base. This equipment-technology dimension of the dependency means that even states with access to mineral ore deposits and the capital to invest in processing infrastructure face a severe bottleneck in acquiring the proprietary equipment and process knowledge required to build competitive refining capacity - a bottleneck that China controls and has demonstrated a willingness to restrict. On 9 October 2025, China announced major export controls on lithium-ion battery supply chains covering battery cells and packs for high-performance applications, cathode precursors, an expanded scope of anode materials, and battery and material production equipment and technologies. The inclusion of manufacturing equipment - not merely final products or raw materials - in the export control regime is the most structurally significant dimension of the October 2025 measures,

because it explicitly targets the capacity of non-Chinese actors to replicate the refining architecture, not merely to source its outputs.

The correct analytical analogy is not to oil production - where the relevant dependency is on resource extraction - but to oil refining in the 1950s and 1960s, when the Seven Sisters oil companies controlled the downstream processing infrastructure through which globally sourced crude was converted into commercially usable products. The power that refinery control conferred in that era was substantially greater than mining or drilling dominance, because it was the transformation stage, not the extraction stage, that determined final product availability and price. China's construction of equivalent dominance in critical mineral refining is a deliberate and long-run strategic achievement that has been decades in the making, and it confers a qualitatively different and more durable form of leverage than simple commodity production concentration.

The 2010 Rare Earth Embargo Against Japan - The Proof of Weaponization Concept

The earliest and most instructive case study of critical mineral supply chain weaponization is China's effective restriction of rare earth exports to Japan in September 2010, following a diplomatic dispute over the detention of a Chinese fishing trawler captain in waters claimed by both countries near the Senkaku/Diaoyu islands. China, at that time controlling approximately 97% of global rare earth production, halted rare earth export approvals to Japan - without formally announcing an embargo, and without reference to the underlying diplomatic dispute in official communications - for a period of approximately six to eight weeks. The operational impact on Japanese industry was significant: rare earth elements including neodymium, dysprosium, and terbium are essential inputs for permanent magnet motors used in hybrid vehicles (Japan's signature automotive sector), precision electronics, and defense applications. Japanese manufacturers experienced supply disruptions and significant price

increases; rare earth prices spiked internationally, with cerium oxide prices increasing by over 500% and dysprosium prices increasing by over 1,000% over the following two years.

The 2010 episode is analytically foundational for several reasons beyond the immediate supply disruption. First, it demonstrated that China was willing to weaponize mineral supply chains in a bilateral dispute entirely unrelated to minerals trade - that is, that the weaponization potential was not contingent on a mineral-specific grievance but was a general-purpose instrument of coercive diplomacy applicable to any political dispute in which China chose to deploy it. China first weaponized rare earths in 2010 when it banned exports to Japan over a fishing trawler dispute - a precedent that established the template for subsequent export control escalations between 2023 and 2025. Second, the episode triggered a significant and sustained Japanese government response - including the establishment of the Critical Materials Strategy, accelerated investment in alternative rare earth sourcing, and a rare earth recycling program - but did not succeed in substantially reducing Japanese rare earth dependency on China over the subsequent decade, demonstrating the structural difficulty of diversification even when the political motivation and government resources are present. Third, and most consequentially, the episode demonstrated that the WTO rules-based trading system - under which Japan pursued a formal complaint, ultimately decided in Japan's favor in 2014 - provided inadequate protection against short-term weaponization: by the time the WTO panel ruled against China's export restrictions, the political dispute had long been resolved and the restrictions had been formally lifted, leaving the finding with limited practical remedy.

Existing Frameworks and Their Structural Limitations

The policy response to critical mineral supply chain risk has generated a proliferation of bilateral and minilateral frameworks over the past four years that are individually promising but collectively

insufficient. Both the US and EU have independently developed economic engagement mechanisms to expand their presence in African mining sectors. The April 2026 EU-US MoU on critical minerals creates potential for these previously separate approaches to become coordinated, though the transition from parallel to joint strategy involves significant institutional complexity. The US-led Minerals Security Partnership (MSP), launched in 2022 and comprising 14 countries plus the EU, has supported the identification and development of critical mineral projects outside China but has been limited in its capacity to compete with Chinese state-backed investment by the political and commercial constraints on partner governments' ability to deploy comparable levels of subsidized capital.

The April 2026 EU-US Critical Minerals Partnership - built around a memorandum of understanding and an Action Plan covering the full supply chain from exploration through recycling - is best understood as a political and institutional bridge rather than a final legal solution. It does not resolve the question of whether EU-sourced critical minerals qualify under the US Inflation Reduction Act's clean-vehicle credit regime, which would require a separate formal agreement. Until such an arrangement is concluded, European industry does not automatically gain full IRA-equivalent status through the April 2026 partnership alone. This gap between the partnership's political ambition and its legal-operational content illustrates the structural deficiency of the current approach: bilateral and minilateral frameworks generate political momentum and coordination precedents, but they do not create the legally binding, commercially operative structures that would redirect capital flows at the scale required.

The United States has pursued an integrated "mine-to-magnet" strategy combining IRA tax credits, Defense Production Act guarantees, equity participation, and long-term off-take contracts - in 2025 alone, the US government took equity stakes in mining and smelting companies, including a 15% stake in MP Materials. This represents the most operationally substantive step taken by any G20 member to

directly address the commercial non-viability problem created by Chinese price suppression, but it is a unilateral national industrial policy instrument rather than a coordinated multilateral response.

5. *PILLAR II: SUPPLY CHAIN RESILIENCE*

The contemporary global economy is organised around global value chains, in which the production of a single finished good is fragmented across many countries, each performing the task in which it holds comparative advantage. This fragmentation is the engine of the productivity gains of recent decades, and it rests on two assumptions that regional instability falsifies. The first is that goods and components can move across borders cheaply, quickly, and predictably. The second is that the cheapest available supplier can always be reached. The dominant inventory philosophy of the integrated era, just-in-time production, deliberately minimised held stock in order to reduce the cost of capital tied up in inventory, on the assumption that resupply would be reliable. Just-in-time is the inventory expression of the option-selling logic described above: it maximises efficiency by holding no buffer, which is optimal until the moment resupply fails, at which point the absence of buffer converts a delay into a stoppage. The post-pandemic and post-2022 reassessment have been a partial and uneven shift back toward just-in-case, the holding of strategic inventory and redundant suppliers, and the tension between the efficiency of the former and the resilience of the latter is the central design problem of supply-chain policy.

Case Study: The Red Sea and Suez Disruption (2023 to 2026)

The disruption of the Red Sea corridor between 2023 and 2026 represents one of the clearest contemporary examples of how regional instability can generate global economic consequences through supply chain networks. Unlike conventional commodity shocks, where disruption originates from a reduction in

production, the Red Sea crisis demonstrated the vulnerability of the global economy to disruptions in the movement of goods itself. The Bab al-Mandab Strait and Suez Canal together form one of the world's most critical maritime chokepoints, linking Asian manufacturing centres with European consumer markets. Approximately 12–15% of global maritime trade and nearly one-third of intercontinental container traffic normally transit through this corridor, making it a vital component of contemporary globalization.

From a G20 perspective, the significance of the crisis lies not merely in the attacks on commercial vessels but in the exposure of a deeper structural vulnerability. Global supply chains over the past three decades have been optimized primarily for efficiency, minimizing transportation costs, inventory holdings, and production redundancies. This model assumes uninterrupted access to major trade corridors. The Red Sea crisis challenged this assumption by demonstrating that a relatively localized security threat could disrupt supply chains connecting dozens of economies simultaneously. When major shipping companies diverted vessels around the Cape of Good Hope, voyage times between Asia and Europe increased significantly, transportation costs surged, and firms were forced to absorb higher logistical expenses throughout their production networks.

More importantly, the crisis revealed a fundamental distinction between efficiency and resilience. An efficient supply chain minimizes costs under normal conditions; a resilient supply chain maintains functionality under adverse conditions. The global trading system proved capable of absorbing the shock primarily because the disruption occurred during a period characterized by spare shipping capacity, relatively low consumer demand, and manageable inventory levels. Consequently, despite dramatic increases in freight costs and shipping delays, the broader macroeconomic impact remained relatively

contained. However, this outcome should not be interpreted as evidence of systemic strength. Rather, it suggests that resilience was provided by favourable market conditions rather than by deliberate policy design.

For G20 economies, this raises a critical strategic question: how resilient would the global economy be if a similar disruption occurred during a period of supply-side tightness? Had the crisis coincided with strong post-pandemic demand, widespread port congestion, or simultaneous disruptions in other strategic corridors, the resulting inflationary and growth effects would likely have been substantially greater. The Red Sea episode therefore highlights that supply chain resilience cannot be measured solely by a system's performance during a single crisis; it must also account for its ability to withstand multiple, overlapping shocks.

The crisis further illustrates the growing interconnection between security policy and economic governance. Traditionally, maritime security has been viewed as a defence issue, while trade facilitation has been treated as an economic matter. The Red Sea disruption demonstrates that these domains are increasingly inseparable. A non-state actor operating in a regional conflict zone was able to influence global freight markets, affect inflationary pressures in major economies, reduce revenues for Egypt, and alter the logistical calculations of multinational corporations. In effect, regional instability became a direct variable in global economic performance.

Consequently, the primary lesson for the G20 is not that supply chains require complete restructuring, but that resilience must become a central objective of economic governance. This includes diversification of transportation routes, greater visibility across supply networks, enhanced protection of strategic maritime

corridors, improved coordination between public and private actors, and the development of contingency mechanisms for critical trade disruptions. The Red Sea crisis demonstrated that the vulnerability of modern supply chains often lies not in production itself but in the narrow corridors through which global commerce flows.

Ultimately, the Red Sea disruption should be understood as a warning against complacency. The crisis did not trigger a global economic emergency because sufficient slack existed elsewhere in the system. Yet reliance on excess capacity is not a resilience strategy. For the G20, the challenge moving forward is to institutionalize resilience before future disruptions occur, particularly as geopolitical competition, regional conflicts, and maritime insecurity become increasingly persistent features of the international system. The central question is therefore no longer whether global supply chains are vulnerable, but whether the international community can adapt quickly enough to prevent localized instability from generating systemic economic consequences.

Single Points of Failure: Semiconductors and Concentrated Inputs

The semiconductor industry represents one of the clearest examples of a systemic supply-chain vulnerability in the contemporary global economy. Unlike energy or shipping shocks, which disrupt the movement of an already-existing commodity, semiconductor disruption strikes the productive core of the global economy itself. Advanced chips are not ordinary inputs: they are embedded in automobiles, smartphones, data centres, telecom networks, medical equipment, industrial automation, and military systems. This makes them a foundational enabling technology rather than a narrow sectoral product. The key structural problem is not merely demand, but extreme concentration. Leading-edge chip fabrication is highly concentrated in a small number of firms and locations, with Taiwan occupying the most sensitive position in the global value chain. That means a disruption in

one place can propagate through dozens of downstream sectors at once.

From a G20 perspective, this is precisely the kind of vulnerability that turns a commercial supply chain into a macroeconomic and geopolitical risk. The problem is not only that Taiwan is concentrated, but that the concentration sits in a politically contested environment. The semiconductor chain has become a single point of failure because the world has built a distributed demand structure on top of a highly concentrated production base. In practice, this means that a blockade, military escalation, cyberattack, or even a natural disaster affecting fabrication capacity could not be rapidly compensated for elsewhere. The shortage would not simply raise prices; it would interrupt production schedules across the entire industrial economy. The consequence is a chain reaction: output losses, delays in manufacturing, inflationary pressure, higher inventory hoarding, and strategic insecurity in sectors that depend on precision chips. The semiconductor case therefore shows that globalization has maximized efficiency, but at the cost of concentrating fragility in a few nodes whose failure would have disproportionate global consequences.

The first flaw is geographic concentration. Advanced fabrication is not evenly distributed, and the most sophisticated nodes are even more concentrated than the broader chip industry. This means the system lacks true redundancy. In a resilient network, multiple suppliers should be able to absorb shocks; in semiconductors, the most advanced production cannot be substituted quickly because the technical frontier is not interchangeable with older generation chips. Older chips can be made in more places, but the highest-end chips require the most advanced facilities, expertise, and equipment. That is why a shock to leading-edge fabs would create a much deeper disruption than a shock to a commodity market.

The second flaw is capital intensity and time delay. A chip fabrication plant is extraordinarily expensive and takes years to bring online. Even if governments decide to diversify now, the supply response is slow because semiconductor ecosystems are not just buildings; they are networks of engineers, clean water, electricity, chemicals, precision tools, logistics, and tacit know-how. The result is that the industry cannot scale up quickly in response to geopolitical risk. In G20 terms, this means resilience is expensive to build and impossible to improvise after a crisis begins.

The third flaw is interdependence across layers of the value chain. Semiconductors are not just designed and fabricated; they are also dependent on a global ecosystem of design software, lithography machines, specialty gases, wafers, chemicals, packaging, testing, and transport. A disruption in one layer can block the rest. This layered structure makes the chain efficient in normal times but brittle under stress, because no single state controls the entire process.

The fourth flaw is inventory logic. Semiconductor supply chains have long been optimized for lean production, just-in-time delivery, and low warehousing costs. That lowers cost, but it also reduces buffer capacity. When demand surges or supply is interrupted, there is little slack in the system. For a G20 committee, this matters because resilience cannot be understood as simply “more production”; it also requires inventory design, better forecasting, and strategic buffering.

The semiconductor issue is not sectoral; it is systemic. A chip shortage cascades through the entire economy because chips are upstream inputs to so many industries. In autos, missing chips mean production delays. In telecoms, they affect network deployment. In defence, they affect procurement and weapons systems. In consumer electronics, they affect product release cycles. In data infrastructure, they affect servers, cloud capacity, and AI systems. That is why a semiconductor shock can behave like a multiplier event: one disruption produces multiple downstream bottlenecks.

For global supply chains, this creates three effects. First, it raises lead times and lowers predictability. Second, it forces companies to hold larger inventories, increasing cost. Third, it intensifies price volatility because supply bottlenecks pass through quickly into downstream markets. For the global macroeconomy, this can feed inflation, lower productivity, and reduce growth. Unlike a local factory shutdown, a semiconductor shock is not confined to one region; it reverberates through the entire international production system.

From the G20's perspective, the semiconductor issue is not simply about making more chips. It is about governing systemic fragility in a fragmented world. The G20 is uniquely relevant because it sits at the intersection of advanced economies, manufacturing powers, and major demand centres. But it also faces a structural dilemma: every member wants resilience, yet not every member wants the same degree of openness, nor the same geopolitical alignment. The central G20 challenge is to avoid turning resilience into decoupling. True resilience means diversification, transparency, and redundancy. Decoupling means splitting the system into rival blocs. The first is expensive but manageable; the second is economically much more damaging. For the G20, the priority should be to reduce single points of failure without creating new political walls around the global economy.

Re-shoring, Friend-Shoring, and Near-shoring

As a response to such vulnerabilities, many countries are restructuring supply chains along geopolitical lines. Reshoring, near-shoring, and friend-shoring all involve relocating production to “safe” partners (home, neighbors, or allied states). In effect, firms pay a resilience premium: they accept higher costs to gain assurance of supply. Recent trade data confirm the trend – advanced economies are importing

relatively more from nearby and politically aligned partners, even if it raises unit costs. Analysts estimate that up to 26% of global exports (about US\$4.6 trillion) could be reallocated under these policies over the next few years. In other words, the world is deliberately “de-globalizing” certain supply lines to buy resilience.

However, this insurance is not free and its costs are unevenly spread. High-income G20 members can afford to pay more for domestic or allied production, but developing economies stand to lose. Critics warn that friend-shoring could effectively exclude poor countries from key markets. As one study notes, it would “exclude poor countries even more strongly from international trade, fail to provide them with income and employment, and endanger their internal stability”. Another analysis observes that cutting out “non-aligned” nations would shrink overall trade and growth. Indeed, an IMF working paper finds that aggressive de-risking (reverting to year-2000 levels of integration) could shrink global GDP by ~4.5% under broad reshoring and by ~1.8% under friend-shoring. Crucially, the paper notes that friend-shoring “does not necessarily deliver a boon to third countries” – any trade gains in one bloc may be offset by losses elsewhere. In practice, then, G20 resilience strategies risk reinforcing economic divides: developing exporters (e.g. in Southeast Asia, Africa, Latin America) may see demand cut as buyers shift to friends.

For G20 deliberations, the challenge is balancing security and inclusion. Policymakers must acknowledge that resilience is costly: doing nothing may appear cheaper in the short run, but relying only on pre-existing slack is dangerous (as the Red Sea case showed). Yet aggressive friend-shoring without a plan risks fragmenting the very global economy the G20 seeks to stabilize. Possible cooperative approaches include helping vulnerable countries diversify into new markets, providing development finance to offset lost demand, or coordinating trade rules so that new “trusted” supply chains remain open to partners that meet high standards (labor, environment, etc.) rather than

geography alone. Ultimately, G20 countries need strategies that enhance supply-chain robustness without permanently sidelining emerging economies. This balance – securing the rich world’s supply lines while upholding an open global trading system – will be the test of whether the G20 can turn supply-chain shocks into an opportunity for *cooperative* governance, rather than deeper fragmentation.

Extreme Geographic Concentration in Extraction and Processing

The mineral supply chain for clean energy technologies exhibits a degree of geographic concentration that, from a supply security perspective, is comparable to - and in several dimensions more severe than - the dependence on Persian Gulf hydrocarbons that has defined energy security for the past five decades. Consider the empirical profile: the Democratic Republic of Congo accounts for approximately 70% of global cobalt production, and China controls approximately 80% of cobalt refining capacity globally. China accounts for approximately 60–65% of global rare earth element production and over 85% of global rare earth processing. Chile and Australia together account for approximately 50% of global lithium production, but China controls approximately 60% of lithium processing capacity. China, Indonesia, and the Philippines together account for approximately 70% of global nickel production. China produces approximately 80% of the world's solar polysilicon, 97% of solar wafers, and between 60–80% of global solar cell and module production capacity. This is not merely a commercial concentration - it is a structural condition in which the decarbonization programs of the advanced G20 economies are, at the current state of supply chain development, critically dependent on the continued willingness of the People's Republic of China to supply processed inputs at commercially viable prices and without the imposition of politically motivated export restrictions. China's imposition of export controls on gallium and germanium in August 2023, on graphite in

December 2023, and on a range of rare earth processing technologies in 2024–2025, demonstrates beyond reasonable doubt that this dependency is recognized in Beijing as a source of leverage and has been operationalized as a strategic instrument.

South Africa's 2025 G20 Presidency and the Critical Minerals Governance Gap

South Africa's presidency of the G20 in 2025 was the first G20 presidency hosted on the African continent, and it placed the governance of critical mineral supply chains - particularly from the perspective of producer nations seeking to capture more value from their mineral endowments - at or near the center of the G20 agenda. The South African G20 agenda advanced the principle of "beneficiation" - the idea that producer nations should not merely export raw mineral ores but should develop domestic processing, refining, and manufacturing capacity that retains more of the value-added component of the supply chain within the producing country. This agenda has both a development economics dimension (processing industries create higher-wage employment and generate more tax revenue than extraction) and a supply security dimension (diversifying processing capacity away from China reduces systemic concentration risk for the global clean energy supply chain). The governance gap that the South African presidency sought to address - but did not resolve - is the absence of any multilateral framework for governing critical mineral supply chains that simultaneously addresses the interests of producer nations (fair value capture, development finance for processing infrastructure), consumer nations (supply security, diversification, price stability), and environmental and social standards (responsible mining practices, community benefit-sharing, environmental remediation). The G20 is the only forum that includes both the major mineral-rich producing nations (South Africa, Indonesia, Brazil, Australia, Argentina) and the major consuming nations (China, the US, EU, Japan, South Korea), and is therefore the natural venue for developing

such a framework.

The G20's engagement with critical minerals has developed substantially since 2021, with the establishment of supply chain resilience working groups, endorsement of the OECD Due Diligence Guidance for Responsible Mineral Supply Chains, and support for multilateral initiatives such as the Minerals Security Partnership (MSP), launched by the United States in 2022 and involving 14 countries plus the EU. However, the MSP and similar "friendshoring" arrangements - discussed further - risk fracturing the global minerals market into aligned and non-aligned blocs in ways that increase aggregate costs, reduce efficiency, and exclude Global South producers who lack the political alignment to participate in Western-led partnerships. The G20's comparative advantage in this domain is precisely its membership breadth: it can pursue a more genuinely multilateral approach to critical minerals governance than either the MSP (too narrow, politically defined) or the WTO (insufficiently mandated for resource governance) can achieve.

Goeconomic Fragmentation, Friendshoring, and the Architecture of the Post-Liberal Trading Order

The concept of "geoeconomic fragmentation" - the progressive decoupling of the global trading, financial, and technological system along geopolitical lines - has moved from the analytical periphery to the analytical center of international economic policy in the period since 2016, and has accelerated dramatically since 2022. Its most precise articulation in policy practice is the doctrine of "friendshoring" - the strategic reorientation of supply chains and investment flows toward countries that share political values and security alignments, explicitly at the expense of economic efficiency considerations. Friendshoring is the supply chain expression of a broader ideological proposition: that the post-Cold War model of economic interdependence as a source of mutual constraint and peace is not merely incomplete but dangerously naïve, and that the economic system must be redesigned to

reflect the geopolitical realities of great-power competition. The G20 is both the primary institutional forum in which this fragmentation is being negotiated and a partial casualty of it - geopolitical divisions have increasingly impeded the consensus-based decision-making that the forum depends on, generating bloc-based positioning and communiqué language so hedged as to be analytically meaningless. Understanding the mechanics, extent, and consequence of geoeconomic fragmentation is therefore both a subject for G20 deliberation and a precondition for understanding what the G20 can realistically achieve.

The economic costs of geoeconomic fragmentation have been quantified with increasing precision by the IMF, World Bank, and academic economists. A 2023 IMF staff analysis estimated that a severe fragmentation scenario - in which the global economy splits into two technology and trade blocs - could reduce global GDP by between 2.5% and 7% in the long run, with the largest losses borne by smaller emerging market economies that lack the scale to maintain competitive domestic supply chains across all strategic sectors and are forced to choose between blocs at significant economic cost. The WTO has estimated that a decoupling of global goods trade along geopolitical lines could reduce trade volumes by 40–50% over the medium to long term. The sectors in which fragmentation is most advanced and consequential include: semiconductors and advanced computing (with US export controls through CHIPS and allied export control regimes creating a bifurcated supply chain); clean energy technology manufacturing (with US IRA domestic content requirements and EU Green Deal industrial policy creating incentives for supply chain localization that discriminate against non-allied suppliers); artificial intelligence and data (with increasingly divergent regulatory frameworks and data localization requirements creating incompatible digital ecosystems); and, following the 2026 Iran crisis, energy supply chains more broadly. The World Economic Forum's 2026 Global Cooperation Barometer notes that while multilateral forms of cooperation have declined, smaller and more agile

coalitions of countries have been instrumental in maintaining overall cooperation levels - a finding that accurately describes the dynamics of friendshoring, but that should be read analytically as a description of fragmentation rather than as a normative endorsement of it.

The G20's capacity to address geoeconomic fragmentation is constrained by the fact that the primary agents of fragmentation - the United States and China - are both G20 members, and their strategic competition is a constitutive feature of the current geopolitical order rather than a problem amenable to G20 resolution. However, it can function as a transparency and norm-setting body - developing agreed frameworks for the notification, justification, and sunset-clause requirements for trade-restrictive measures taken on national security grounds, under the WTO's Article XXI security exception, which lacks procedural discipline.

6. PILLAR III: MACROECONOMIC STABILITY

The first two pillars describe physical disruptions; this pillar describes how those disruptions become macroeconomic events. Three channels operate simultaneously. The price channel transmits a supply disruption into higher input costs, which propagate through the economy as inflation. The output channel transmits a disruption into reduced production, as firms unable to obtain energy or components curtail output, which lowers growth and employment. The expectations channel, the most insidious and the hardest to measure, transmits the anticipation of future disruption into present behaviour: firms hoard, investors demand higher returns to compensate for elevated risk, and households and governments build precautionary buffers. The expectations channel is the macroeconomic expression of the option-repricing mechanism, and it explains why economic damage frequently precedes and exceeds any physical disruption that actually materialises.

Stagflationary Risk, Monetary Policy Dilemmas, and the Limits of Central Bank Coordination Under Geopolitical Shock

Stagflation - the simultaneous occurrence of elevated inflation, stagnant or negative economic growth, and rising unemployment - occupies a special place in macroeconomic history because it is the condition that definitively exposed the limitations of the Keynesian demand-management framework that governed Western economic policy for the postwar period, catalyzing the monetarist revolution of the late 1970s and establishing the intellectual and institutional foundations of the inflation-targeting, central bank independence paradigm that has governed monetary policy in advanced G20 economies for the past four decades. The return of stagflationary conditions - or the credible risk thereof - in the wake of the 2026 energy shock is therefore not merely an economic policy problem; it is a test of the conceptual adequacy of the macroeconomic governance framework itself. Stagflation produced by an energy supply shock presents a uniquely difficult monetary policy challenge because the inflationary and contractionary effects arrive simultaneously and originate from the same external source - meaning that the standard central bank toolkit of interest rate adjustment addresses one symptom at the cost of aggravating the other. The G20 was created in response to a financial crisis; it must now demonstrate that its macroeconomic coordination architecture is equally capable of functioning in response to a geopolitically generated supply-side shock of this nature and magnitude.

Standard monetary policy theory, as developed in the New Keynesian framework that underlies the inflation-targeting regimes of the Federal Reserve, the European Central Bank, the Bank of England, and equivalent institutions in other G20 economies, is structured around the manipulation of aggregate demand to maintain price stability. When inflation is elevated, the central bank raises interest rates to reduce borrowing and spending, cooling demand and relieving upward price pressure. When growth

is weak, the central bank cuts interest rates to stimulate borrowing and investment. The two instruments are in principle operated on a single dimension - the policy interest rate - and their application is relatively straightforward when inflation and growth move in opposite directions. Stagflation, however, places them in direct conflict: raising interest rates to fight supply-shock-driven inflation chokes investment and consumption in an economy already struggling with high energy costs and weakening demand. Cutting interest rates or holding them steady to support growth risks entrenching inflationary expectations at levels that, once de-anchored, are extremely costly to restore - as the Volcker disinflation of 1980–82, which required interest rates of over 20% and a deliberate recession, demonstrated. The OECD's March 2026 Economic Outlook projected G20 headline inflation at 4.0% for 2026, substantially above the 2% targets of major central banks, while simultaneously projecting global GDP growth of 2.9% - below the 3.5% trend rate of the 2010s. These projections assume that "the most acute disruptions end in May" and that Hormuz flows return to near-normal by late 2026. If the disruption persists beyond these assumptions - which remains a significant risk given the unresolved nature of the Iran-US conflict as of the date of this analysis - the stagflationary pressure intensifies and the growth forecast deteriorates further.

The G20's Mutual Assessment Process (MAP), established in 2009 as a framework for G20 members to assess the collective consistency of their macroeconomic policies, provides the institutional vehicle for the kind of coordinated policy response that the stagflation risk demands. However, the MAP has consistently underperformed its potential, partly because the macroeconomic policy frameworks of the major economies diverge in analytically significant ways - the US Federal Reserve's dual mandate (inflation and employment) differs structurally from the ECB's singular price stability mandate, creating systematic differences in how the two institutions weight the stagflationary trade-off - and partly because the political economy of domestic monetary policy makes genuine external

accountability extremely difficult to sustain. The G20 should not attempt to harmonize central bank mandates - that would be both technically inappropriate and politically unachievable - but it can strengthen the MAP framework in several directions: requiring more granular disclosure of the macroeconomic model assumptions underlying national policy projections, developing a G20 Energy Shock Impact Assessment Protocol that produces a common analytical baseline for the distributional effects of the energy shock across member economies, and exploring the scope for coordinated fiscal policy responses - particularly targeted energy price support for vulnerable households and small businesses - that reduce the demand for monetary accommodation without sacrificing distributional equity.

Sovereign Debt Distress, the Food-Energy-Finance Nexus, and the Macroeconomic Vulnerability of the Global South.

The macroeconomic impact of regional instability is not uniformly distributed across the global economic system. While the immediate political focus of the 2026 energy shock has concentrated on the price exposure of G20 consumer economies - the United States, Europe, Japan, China, India - the most severe and durable macroeconomic damage is accruing to a set of economies that exist at the intersection of multiple compounding vulnerabilities: high import dependence on both energy and food, elevated sovereign debt-to-GDP ratios incurred through pandemic-era fiscal expansion, limited access to external financing at acceptable rates, currency weakness relative to the dollar in which most energy and food commodities are priced, and fiscal space that has already been substantially exhausted. These are the economies of the Global South - a diverse and heterogeneous set that includes sub-Saharan African states, South and Southeast Asian developing countries, and Latin American and

Caribbean economies - and their experience of the 2026 energy shock is analytically and politically distinct from that of the advanced G20 economies in ways that the committee's analysis must comprehend with precision. The food-energy-finance nexus - the structural linkage between energy prices (which determine fertilizer costs, irrigation costs, and food transport costs), food prices (which represent a large share of household expenditure in low-income economies), and financial stability (which determines the cost and availability of sovereign borrowing to finance the resulting fiscal deficits) - is the analytical framework within which the Global South's macroeconomic vulnerability must be assessed.

The accumulation of debt in Lower-Income Countries (LICs) is a complex phenomenon driven by a combination of fiscal policy choices and significant Stock-Flow Adjustments (SFAs). Research consistently indicates that both these elements are critical determinants of debt dynamics in LICs. SFAs, in particular, play an overwhelming role in debt accumulation across all debt surge brackets in LICs, often explaining more than half of these adjustments. A primary contributor to SFAs is the valuation effect stemming from currency depreciation, which is especially pronounced for countries with a high proportion of external debt denominated in foreign currencies. When a local currency depreciates, the domestic value of foreign currency-denominated debt automatically increases, even without new borrowing, significantly inflating the debt burden.

Beyond currency fluctuations, SFAs are also significantly influenced by other factors. These include the accumulation of arrears (unpaid obligations), revisions in debt statistics that reveal previously unrecorded liabilities, the recapitalization of distressed banks and State-Owned Enterprises (SOEs), off-budget transactions that bypass conventional fiscal oversight, and the uncovering of previously hidden debt. SOE-related losses and contingent liabilities, in particular, represent a substantial and frequently underestimated driver of debt accumulation in many LICs. These non-valuation related SFAs tend to be "sticky," meaning they significantly increase the probability of debt remaining

permanently higher.

Compounding these SFA-driven dynamics is the persistent running of primary deficits. LICs have consistently operated with primary deficits, averaging 2.3% of GDP between 2011 and 2023, which has been a primary contributor to the buildup of government debt. This fiscal weakness is often exacerbated by underlying structural issues. LICs frequently collect less than two-thirds of their potential tax revenue, severely constraining their government spending capacity. Furthermore, public spending efficiency is often suboptimal, with a disproportionately large share of expenditures being allocated to interest payments on existing debt, poorly targeted subsidies, and an inflated government wage bill, rather than productive, growth-enhancing investments. The combination of these factors creates a precarious financial situation, where debt surges in LICs are estimated to result in a financial crisis with a probability ranging from 11% to 20%. This underscores the severe macroeconomic risks inherent in their current debt trajectories.

The analysis of these debt drivers reveals a critical underlying issue: the pervasive influence of hidden drivers and systemic opacity within the financial management of LICs. The data clearly indicates that Stock-Flow Adjustments (SFAs), particularly those arising from currency depreciation and previously "hidden debt" (such as liabilities of State-Owned Enterprises and off-budget transactions), are dominant factors in debt accumulation, often overshadowing the impact of explicit fiscal deficits. This observation suggests that the problem of debt in LICs is not solely a matter of government borrowing and spending decisions. The significant role of currency depreciation means that external economic conditions-such as global interest rate hikes, fluctuations in commodity prices, or other exogenous shocks that lead to currency weakening-have a direct, non-fiscal, and often sudden impact on the real value of foreign currency-denominated debt. This implies that a country's debt burden can surge dramatically even without new borrowing or increased domestic spending. Furthermore, the substantial contribution of "hidden debt" from SOEs and off-budget activities points to profound

governance weaknesses and a lack of transparency within national financial management systems. These liabilities are frequently not fully captured in official statistics, creating a systemic opacity that prevents accurate risk assessment and proactive management by both national authorities and international partners. Therefore, any equitable roadmap for debt sustainability must extend beyond traditional fiscal consolidation. It must encompass comprehensive risk management strategies for external vulnerabilities, robust oversight mechanisms for state-owned enterprises, and an unwavering commitment to public debt transparency. Without effectively addressing these "hidden" and externally driven components, efforts to manage debt through fiscal policy alone will remain insufficient and prone to recurrent crises, undermining long-term stability and development.

Assessment of Existing Debt Relief Frameworks

The international community has developed several frameworks to address debt vulnerabilities in LICs, each with its objectives and inherent limitations.

The **IMF-World Bank Debt Sustainability Framework (DSF) for LICs** serves as the primary analytical tool employed by multilateral institutions to assess debt sustainability risks. This framework systematically classifies countries based on their assessed debt-carrying capacity, establishes indicative thresholds for selected debt burden indicators, and evaluates baseline projections against rigorous stress test scenarios. Its core objective is to support LICs in achieving their development goals while simultaneously minimizing their risk of experiencing debt distress. Annual Debt Sustainability Analyses (DSAs) are a mandatory component for IMF surveillance and lending decisions, as well as for determining World Bank IDA credit-grant allocations.

Significant debt relief efforts include the **Highly Indebted Poor Country (HIPC) Initiative** and the **Multilateral Debt Relief Initiative (MDRI)**. Launched by the World Bank and IMF in 1996 and

2005, respectively, these initiatives represented the first substantial international efforts to tackle global debt. Collectively, they led to the cancellation of approximately \$130 billion in debt for 36 countries between 1998 and 2010. A notable success of these initiatives was the significant increase, by around 40%, in poverty-reducing public expenditure in HIPC countries following debt cancellation. However, a critical limitation of both HIPC and MDRI was their failure to establish a robust framework to prevent future debt crises, primarily due to the absence of mechanisms to compel private creditors to participate in debt relief on comparable terms.

In response to the economic fallout of the COVID-19 pandemic, the G20 introduced the **Debt Service Suspension Initiative (DSSI)**. This initiative provided valuable, albeit temporary, breathing space for the poorest countries by pausing official debt payments. Despite its immediate relief, the DSSI's overall effectiveness was constrained by a lack of comprehensive participation from private and multilateral creditors.

Building upon the DSSI, the **G20 Common Framework for Debt Treatments (CF)** was established in late 2020. The CF aimed to provide a structured approach to comprehensive sovereign debt restructurings for LICs, explicitly requiring private creditors to participate on comparable terms to ensure fair burden-sharing. However, despite its ambitious intentions, the implementation of the Common Framework has faced significant challenges. As of December 2021, only three countries—Chad, Ethiopia, and Zambia—had formally requested debt relief under the CF, and each of these cases experienced considerable delays. These delays are attributable to complex coordination issues among diverse creditor groups, including traditional Paris Club members and new creditors such as China and India, as well as country-specific factors. Furthermore, debtor countries have demonstrated hesitancy to participate in the Common Framework, citing concerns about potential damage to their financial market reputation, uncertainty regarding the specific terms and outcomes of the restructuring process, high administrative and negotiation costs, and the prerequisite of entering into an IMF arrangement.

A persistent and critical limitation across all major international debt relief initiatives-HIPC, MDRI, DSSI, and the Common Framework-is the systemic difficulty in ensuring comprehensive and comparable participation from private creditors. This fundamental flaw leads to "too little, too late" debt relief, perpetuates moral hazard, and significantly undermines the overall effectiveness of global efforts. The consistent struggle to secure private creditor involvement on comparable terms means that even when official debt is relieved, a substantial portion of the debt burden, often held by private entities at higher interest rates, remains. This creates a scenario where public funds, sourced from both debtor countries and official creditors, effectively "bail out" private lenders, as vividly illustrated by the cases involving "vulture funds". The absence of a binding, comprehensive, and equitable mechanism for all creditors to participate in debt restructuring allows private creditors to "hold out" for more favorable terms, thereby delaying resolution, increasing the overall cost of the crisis, and forcing debtor countries into deeper distress or repeated, insufficient restructurings. The observed hesitancy of debtor countries to engage with frameworks like the Common Framework, driven by concerns over reputational damage and outcome uncertainty, further compounds this issue. This indicates that the international financial architecture currently lacks a robust, enforceable "bankruptcy" or orderly resolution procedure for sovereign debt. This systemic flaw perpetuates cycles of debt crises, undermines timely and effective relief, and ultimately shifts the burden onto the most vulnerable populations and public finances. A truly equitable roadmap necessitates a fundamental reform of this architecture to ensure universal participation and fair burden-sharing across all creditor types.

Principles of Responsible Lending and Borrowing

The formulation of an equitable roadmap for debt sustainability must be firmly anchored in universally accepted principles of responsible lending and borrowing, recognizing that both sovereign debtors and

their creditors share accountability for preventing unsustainable debt situations. Irresponsible financing practices, whether by borrowers or lenders, can have severe and far-reaching consequences, not only for the debtor country and its citizens but also for its neighbors, trading partners, and the broader international financial system.

A core tenet is **shared responsibility and accountability**. Each party in a sovereign lending transaction-the borrower and the lender-is accountable for its conduct. Neither can wholly shift the duty of ensuring that the agreement is economically beneficial, financially sound, legally authorized, appropriately documented, and carefully monitored.

Lenders bear a significant responsibility for **due diligence and informed decision-making**. They must conduct realistic assessments of a sovereign borrower's capacity to service a loan, based on the best available information and objective technical rules of due diligence and national accounts. Lending beyond a borrower's reasonable repayment capacity not only risks default on the specific loan but also adversely affects the position of all other creditors of that sovereign debtor. Furthermore, lenders should provide comprehensive information to assist borrowers in making informed credit decisions, acknowledging the wide variations in financial sophistication among sovereign entities.

Ensuring **appropriate authorization and legal enforceability** is another critical principle. Lenders are obligated to determine, to the best of their ability, that the financing has been appropriately authorized by the sovereign and that the resulting credit agreements are valid and enforceable under relevant jurisdictions. Should a lender determine that these conditions are not met, they should desist from concluding the agreement.

On the borrower's side, **accountability and the protection of public interest** are paramount. Government officials involved in sovereign lending and borrowing transactions act as agents for the State and its citizens, and are therefore responsible for protecting the public interest. Lenders must refrain from any attempts to suborn government officials or engage in corrupt practices, such as

offering bribes.

Finally, enhanced **transparency and data availability** are crucial for both preventing and effectively resolving debt crises. The IMF and World Bank actively foster debt transparency and support countries in strengthening their capacity to report and manage public debt, recognizing that transparency is not only useful for restructuring but also for understanding risks and opportunities.

The emphasis on responsible lending principles, particularly lender due diligence on repayment capacity and proper authorization, directly addresses systemic issues such as "hidden debt" and corruption that have profoundly impacted LICs, as tragically exemplified by Mozambique's hidden debt scandal. The underlying problem is that when lenders fail to rigorously assess a borrower's capacity or overlook red flags regarding authorization and transparency, they actively contribute to the creation of unsustainable debt. This is not merely a "borrower problem" but rather a shared failure of governance and accountability within the global financial system. By effectively enforcing these principles, ideally through an independent "referee" institution as suggested by some analyses, the international community can establish a more robust framework that prevents the accumulation of illegitimate or unsustainable debt from the outset. This implies that responsible lending and borrowing principles are not merely aspirational ethical guidelines but are fundamental to creating a more equitable and stable international financial system. Their effective implementation, supported by strong regulatory oversight and comprehensive transparency mechanisms, is crucial for breaking the cycle of debt accumulation driven by predatory practices or weak governance, thereby fostering genuine sustainable development.

Strengthening Collective Action Mechanisms

Beyond the DSF, the G-20 Common Framework for Debt Treatments requires significant

improvements to enhance its effectiveness. There must be greater clarity on the different steps and timelines within the Common Framework process to accelerate decision-making, coupled with earlier engagement of both official and private creditors. A comprehensive and sustained debt service payment standstill for the duration of negotiations would provide crucial relief to the debtor country during periods of stress and incentivize faster restructuring procedures. Furthermore, the framework must clarify how comparability of treatment will be effectively enforced across all creditor types, potentially through the rigorous implementation of IMF arrears policies, to provide comfort to both creditors and debtors and ensure fair burden-sharing. The Common Framework should also be expanded to include other highly-indebted countries that could benefit from coordinated debt treatment, moving beyond the current limited scope. The Global Sovereign Debt Roundtable, jointly established by the IMF, World Bank, and G-20, provides a valuable platform for improving creditor-debtor dialogue and addressing restructuring challenges, and its role should be strengthened.

The Potential Role of a UN Framework Convention on Sovereign Debt

A critical gap in the existing international financial architecture is the lack of a standing approach and a dedicated body to preserve the institutional memory of sovereign debt episodes and facilitate orderly resolution. The absence of a robust, enforceable "bankruptcy" or orderly resolution procedure for sovereign debt perpetuates cycles of crises and leads to "too little, too late" relief. A UN Framework Convention on Sovereign Debt could address this fundamental deficiency by establishing an independent debt restructuring process. This process would be based on debt sustainability analyses developed independently from both debtors and creditors, thereby directly addressing the inherent power imbalance and conflict of interest that often compromise the credibility and effectiveness of

current debt resolution efforts. Such a framework would provide clear rules and procedures, diminishing risks for all parties and eliminating incentives for "vulture funds" to exploit distressed situations.

The G20's engagement with developing country debt distress is primarily conducted through the Common Framework for Debt Treatments, launched at the November 2020 Riyadh Summit as a G20-Paris Club mechanism for coordinating official bilateral creditor debt relief for low-income countries eligible for the Debt Service Suspension Initiative. The Common Framework has been widely criticized for its slowness, its exclusion of private creditors and multilateral development bank debt, and its failure to provide timely and sufficient relief in the cases of Zambia, Ethiopia, Ghana, and Sri Lanka - the first countries to undergo Common Framework processes. A central structural deficiency of the Common Framework is the participation problem: China, as the world's largest bilateral creditor to developing countries through its Belt and Road Initiative lending, has been a member of the Common Framework but has consistently applied a more conservative interpretation of comparability of treatment with Paris Club creditors, slowing and in several cases effectively blocking the timely completion of debt restructuring agreements. The G20's US Treasury Secretary Scott Bessent, at the April 2026 Spring Meetings, explicitly called for "coordinated action to ensure fertilizer access amid disruptions in food trade supply chains caused by the war in the Middle East" - a formulation that acknowledges the food-energy-finance nexus but frames the response primarily in supply chain terms rather than in the debt relief and fiscal space terms that developing country governments themselves have consistently identified as the primary constraint on their crisis response capacity. The G20 should pursue a more comprehensive response framework that treats the energy shock's impact on developing countries as a compound crisis requiring simultaneous action on debt service relief (through acceleration and deepening of Common Framework processes),

concessional energy procurement facilitation (through multilateral development bank intermediation), and food security financing (through the World Food Programme and the Global Alliance for Food Security).

The Critical Role of Infrastructure for Sustainable Development

Strategic infrastructure is unequivocally a fundamental driver in the global fight against poverty and a cornerstone for sustainable development. This encompasses a broad range of essential services, including robust transport networks, reliable energy systems, access to clean water and sanitation, and pervasive digital connectivity. These services directly enhance people's connections to jobs and economic opportunities, improve the availability and affordability of goods and services, and fundamentally underpin productivity and overall economic growth. Empirical studies consistently demonstrate a positive correlation between the stock of infrastructure assets and the rate of economic growth, with telecommunications, roads, and electricity networks exhibiting the most significant impacts. As countries transition from primary economic industries to more diversified secondary and tertiary sectors, the quality and availability of infrastructure become even more critical for sustained progress.

The rapid pace of global urbanization, with the majority of projected population growth occurring in developing countries and urban settlements, further underscores the necessity of substantial infrastructure investment. Adequate urban infrastructure is essential not only to accommodate burgeoning populations but also to ensure sustainable urban development, manage congestion, and provide basic services to all residents.

Beyond economic growth and urbanization, infrastructure investment is crucial for achieving broader development, inclusion, and environmental objectives. This includes designing infrastructure to ensure

lower emissions, higher efficiency, and enhanced resilience to climate change. It is estimated that between 10% and 15% of the required infrastructure investment should be specifically allocated to making it sustainable. While such investments may entail additional upfront costs, their net economic effect is strongly positive, yielding significant benefits through improved efficiency, enhanced energy security, greater safety, cleaner methods, biodiversity preservation, technological discovery, and fundamentally reduced climate risk. Long-lived infrastructure assets, which can last up to 100 years, play a major role in determining societies' and economies' readiness for future environmental challenges.

Furthermore, as global trade assumes an increasingly vital role in countries' development trajectories, robust and efficient infrastructure is essential for seamless integration into the global economy. This includes traditional transport infrastructure such as roads, railways, and ports, and increasingly, modern information technology infrastructure like broadband networks. These networks facilitate better integration of supply chains and enable international trade in services, such as outsourcing. As developing countries expand their manufacturing and services sectors, the intensity and quality of infrastructure become paramount to leverage network externalities and ensure competitiveness.

The increasing emphasis on "sustainable infrastructure" and the specific allocation for climate resilience signifies a critical evolution in the understanding of developmental infrastructure. This perspective moves beyond merely addressing quantitative deficits in infrastructure provision to prioritizing the quality, long-term resilience, and environmental impact of projects. The explicit focus on ensuring "lower-emissions, higher efficiency and resilience to climate change" implies that future infrastructure investments must be "future-proofed" against the escalating impacts of climate change and actively contribute to decarbonization, rather than simply replicating past, often environmentally damaging, models. Investing in green and resilient infrastructure upfront, although potentially incurring higher initial costs, generates substantial long-term economic benefits, reduces future

climate-related damages, and aligns with broader global sustainability goals. Therefore, "developmental infrastructure" in the 21st century must inherently be "climate-smart" and resilient. This necessitates not only innovative financing mechanisms but also new technical expertise and a fundamental shift in planning paradigms to ensure that infrastructure serves as a robust foundation for sustainable, inclusive growth, rather than inadvertently creating new vulnerabilities or exacerbating environmental challenges.

Challenges and Inefficiencies in Infrastructure Provision in LICs

Despite the critical importance of infrastructure, Lower-Income Countries (LICs) face pervasive challenges and inefficiencies in its provision, often undermining the development impact of investments.

Firstly, **pervasive deficits and inadequate services** remain a critical issue. Despite significant investments over the past decades, the quantity and quality of infrastructure services in most developing countries remain critically low, falling well below demand. Billions of people still lack access to basic services, and existing networks frequently suffer from deterioration and unreliability. This inadequacy leads to congestion, service rationing, and unserved areas, directly impacting quality of life and economic productivity.

Secondly, **inefficiency and waste** are widespread. A substantial portion of resources allocated to infrastructure is lost due to inefficiencies, including poor operational performance in sectors like power utilities and ports, and general waste throughout the project lifecycle. Inadequate maintenance is an almost universal and costly failure, leading to the rapid deterioration of assets and necessitating much larger expenditures on rehabilitation or replacement in the long term. Budgetary austerity often exacerbates this by leading to cuts in maintenance expenditure, creating a false economy.

Thirdly, **poor planning and project selection** frequently undermine infrastructure investments. Many countries suffer from misallocated project investments, resulting in the construction of inappropriate infrastructure or the provision of services at the wrong standard. Common reasons for project failure include unclear objectives, unrealistic targets, inadequate risk evaluation, and poor forecasting of demand. Politically motivated projects often override considerations of economic viability, leading to "white elephant" projects that fail to generate sufficient income or returns.

Fourthly, **weak management, governance, and corruption** are major systemic issues. Agency problems, weak institutional environments, and corruption are primary reasons why infrastructure projects frequently fail to meet their timelines, budgets, and service delivery goals. The lack of managerial and financial autonomy for service providers, coupled with insufficient accountability mechanisms, exacerbates these inefficiencies.

Finally, **fiscal and absorptive capacity constraints** limit the scale and pace of infrastructure development. LICs often face limited fiscal space, meaning their budgets are already strained by existing obligations. Furthermore, they may lack the absorptive capacity-the technical and managerial resources-to effectively implement numerous large-scale projects simultaneously, even when sufficient financing is available. Studies suggest that when public investment exceeds a certain percentage of GDP, the productivity of investment projects may decline due to these constraints.

Case studies of infrastructure project failures vividly illustrate these challenges:

- **Ghana-STX Building Project:** This \$10 billion housing project failed due to haphazard management, disunity among partners, alleged corruption, and a fundamental lack of credible feasibility studies and proper planning. The project's abandonment resulted in significant financial losses.
- **Egypt's Toshka, New Valley Project:** This ambitious \$90 billion agricultural development project, aimed at creating a "second Nile Valley," largely failed because critical environmental

factors, such as high saline levels and underground aquifers, were not adequately considered during planning. Additionally, there was insufficient stakeholder consultation, leading to unforeseen problems.

- **African Water Projects (Boreholes):** Hundreds of millions of dollars were reportedly wasted on water projects across rural Africa, with approximately 50,000 water supply points becoming non-functional. The causes included poor construction, a lack of maintenance expertise, inadequate supervision, insufficient long-term support, and crucially, a lack of consultation with local communities regarding the costs and responsibilities of ongoing maintenance and ownership. This highlights a disconnect between project design and local realities.
- **Sri Lanka's Hambantota Port and Mattala Rajapaksa International Airport:** These large-scale, Chinese-financed projects became unsustainable due to a combination of political motivations and unsound project selection. They were built in underdeveloped areas without sufficient economic justification, driven by the former president's aim to transform his hometown. The opaque, state-to-state nature of the Belt and Road Initiative (BRI) project selection further contributed to their failure to generate sufficient income, leading to billions in additional debt obligations. The IMF estimates that 30% of potential benefits from public investment in emerging markets are lost due to inefficient planning, implementation, and corruption.

The consistent pattern of infrastructure project failures in LICs, as evidenced by these case studies, reveals that the fundamental constraint is often a deficit in governance, planning, and institutional capacity, rather than merely a lack of financing. The observation that projects frequently fail due to poor planning, weak management, corruption, inadequate maintenance, and political interference suggests that simply injecting more capital into these systems will not resolve the underlying problems.

The root causes of these failures lie in systemic weaknesses within public investment management

(PIM) frameworks, including inadequate appraisal, selection processes vulnerable to political capture, and insufficient oversight during implementation and operation. For instance, Sri Lanka's "white elephant" projects demonstrate how political motivations can override economic viability, leading to the creation of debt-creating assets that fail to generate returns. Similarly, the widespread failure of water projects in Africa due to a lack of maintenance and local consultation highlights a profound disconnect between project design and local realities and capacities. This implies that simply increasing financial flows for infrastructure without simultaneously addressing these underlying governance and institutional weaknesses will inevitably lead to continued project failures, wasted resources, and further debt accumulation without commensurate development impact. An equitable roadmap must therefore prioritize strengthening public investment management, enhancing transparency and accountability, and fostering genuine local ownership and capacity throughout the entire project lifecycle.

Innovative Financing Mechanisms for Infrastructure

Addressing the vast infrastructure deficits in Lower-Income Countries (LICs) necessitates a diversified approach to financing, leveraging innovative mechanisms that can mobilize significant capital while ensuring sustainability.

Public-Private Partnerships (PPPs)

Public-Private Partnerships (PPPs) offer a viable strategy for financing and managing infrastructure by strategically leveraging private sector investment, expertise, and innovation. The benefits of PPPs are multifaceted: they can mobilize additional funding sources for infrastructure projects, especially when governments face financial constraints. By requiring projects to attract private finance, PPPs can enhance the project selection process, ensuring that more viable and well-vetted projects are

undertaken. PPP contracts are designed to align the private sector's incentives with public interests throughout the project's entire life cycle, including implementation, by linking private operator revenue to pre-agreed performance indicators and requiring significant, long-term capital investment. This incentive framework often fosters efficiency gains that can outweigh the additional cost of private finance. Countries with extensive PPP experience have observed that PPPs generally manage construction better than traditional public procurement, with projects more frequently completed on time and within budget. The long-term investment horizon of PPP contracts also helps ensure that assets are maintained in good, serviceable condition over their operational life. Additional benefits include enabling whole-of-life costing, facilitating risk transfer to the party best able to manage it, providing predictability of costs, focusing on service delivery, fostering innovation, optimizing asset utilization for multiple revenue streams, and enhancing accountability through performance-linked payments. Critical success factors for PPPs include strong legal and regulatory frameworks, transparent contracts, and fair risk-sharing arrangements between public and private partners. The World Bank's PPP Resource Center offers extensive guidance and tools for developing such frameworks.

Despite their considerable benefits, PPPs also present significant challenges and limitations. They can involve political risks, financial complexity, regulatory obstacles, and potential public opposition stemming from concerns about privatization. A significant pitfall is the potential for "unclear fiscal commitments" and the tendency to treat PPPs as "off-balance sheet" liabilities. This accounting ambiguity can create a fiscal illusion, leading governments to accept higher fiscal commitments and risks than prudent public financial management would allow. PPPs are not a universal solution for all infrastructure problems and require sound public decision-making based on comprehensive analysis and a robust governance framework promoting transparency and accountability. Specific challenges for serving the poor through PPPs include issues with monopoly providers, legal prohibitions in

informal settlements, the imposition of inappropriate engineering standards, and difficulties related to tariffs and billing mechanisms that may not be affordable or accessible for low-income populations. The tendency to treat PPPs off-balance sheet creates a significant fiscal illusion that can undermine debt sustainability. While PPPs are often lauded for their ability to attract private finance and transfer risks from the public to the private sector, the challenge arises from the fact that their fiscal commitments can be unclear and are sometimes treated as "off-balance sheet." This ambiguity can cause governments to accept "higher fiscal commitments and risks than prudent public financial management would allow". The underlying dynamic is that this accounting practice can mask the true extent of public liabilities and contingent risks associated with PPPs, making them a potential source of future debt distress if not transparently managed. The perceived "free money" aspect or the ability to bypass immediate budgetary scrutiny can incentivize governments to undertake projects that might otherwise be deemed fiscally imprudent. To ensure that PPPs genuinely contribute to sustainable development without creating new debt vulnerabilities, there must be absolute transparency in reporting all PPP-related liabilities. This necessitates robust fiscal risk assessment frameworks, such as the PPP Fiscal Risk Assessment Model (PFRAM), and strong governance to prevent political motivations from overriding economic and fiscal prudence. The G-20 should advocate for international standards that mandate full disclosure and integration of all PPP liabilities into national debt sustainability analyses, thereby ensuring a complete and accurate picture of a country's fiscal health.

Blended Finance

Blended finance is a strategic approach that leverages catalytic capital from public or philanthropic sources to de-risk projects and make them more attractive to private investors, thereby mobilizing private capital for sustainable development. It is considered a vital tool for achieving the Sustainable Development Goals (SDGs) and Nationally Determined Contributions (NDCs) under the Paris

Agreement, particularly in emerging and developing markets. While blended finance has mobilized over \$200 billion in the past decade, this amount remains "far short of what is required to achieve our collective sustainability and net-zero targets" and to bridge the multi-trillion dollar annual financing gap for the SDGs.

Key enablers for the success of blended finance vehicles include:

- **Alignment in Objectives:** All participating stakeholders-public, private, and philanthropic-must have aligned objectives from the outset to prevent structural inefficiencies and facilitate fundraising and capital deployment.
- **Robust & Proven Track Record:** The reputation and credibility of the fund manager are pivotal throughout the vehicle's lifespan, fostering investor confidence and promoting repeat investments.
- **Risk-Profile Enhancing Instruments:** Strategic deployment of instruments such as first-loss capital, insurance/guarantees, and credit ratings is essential to mitigate investor concerns and improve creditworthiness.
- **Local Knowledge & Expertise:** Teams with deep local understanding significantly strengthen the risk management framework and help identify high-potential opportunities aligned with local government priorities.
- **Top-Down Endorsement:** Senior leadership support accelerates market entry, attracts important stakeholders, and secures necessary resources for fund launches.
- **Simple & Flexible Vehicle Design:** A straightforward yet adaptable design expedites stakeholder alignment, lowers development time, and facilitates replicability.

Successful case studies include:

- **SunFunder (Africa):** A leading debt-financing provider for distributed solar projects across Africa and other emerging regions, SunFunder has closed over \$150 million in loans to 57

solar companies. These investments have significantly improved energy access for over 8 million people and mitigated an average of 750,000 tons of CO₂ per year, demonstrating the potential of blended finance in clean energy infrastructure.

- **NWP and KIFFWA Partnership (Kenya):** The Netherlands Water Partnership (NWP) and the Kenya Innovative Finance Facility for Water (KIFFWA) collaborated to provide capital and guidance for climate-resilient water initiatives. This partnership funded 8 water initiatives, providing 30,000 Kenyans with access to potable water at less than one-third of the original cost per liter, showcasing effective blended finance in critical social infrastructure.

The core value proposition of blended finance lies in its ability to strategically deploy catalytic public or philanthropic capital to fundamentally alter the risk-return profile of developmental infrastructure projects, thereby unlocking much larger pools of private capital that would otherwise remain on the sidelines. Private capital is often hesitant to invest in LIC infrastructure due to high perceived risks and unattractive risk-adjusted returns. Blended finance addresses this by using concessional public or philanthropic capital, such as first-loss capital or guarantees, to absorb initial risks or enhance returns for private investors. This mechanism directly addresses the "unattractiveness of risk-adjusted return" by de-risking projects, making them palatable for institutional investors. The success of projects like SunFunder and the NWP/KIFFWA partnership demonstrates that this approach can effectively mobilize private capital for critical sectors like renewable energy and water access. To scale up blended finance effectively, the G-20 should focus on standardizing vehicle structures, strengthening risk mitigation instruments, fostering local knowledge, and providing top-down endorsement. This ensures that catalytic capital genuinely mobilizes new private investment rather than displacing it, and that projects align with national development priorities and climate goals, maximizing their impact.

Green Bonds

Green bonds represent an increasingly important and innovative debt financing instrument specifically designed to channel capital towards projects with demonstrable environmental benefits, such as renewable energy and low-carbon transport infrastructure. The transport sector, with its predictable cash flows and long-term asset nature, holds significant potential for green bond utilization.

However, a key challenge for most cities and governments in developing countries to issue green bonds directly is their insufficient creditworthiness, which limits their independent access to these markets. Consequently, Development Finance Institutions (DFIs) like the World Bank and Asian Development Bank have emerged as major issuers of green bonds in this context, acting as intermediaries to channel funds to eligible projects.

Successful case studies include:

- **IFC Green Bond Program:** Launched in 2010, the International Finance Corporation's (IFC) Green Bond Program aims to catalyze the market and unlock private sector investment for climate-smart projects. In fiscal year 2024, IFC delivered \$19.4 billion in climate finance through this program, supporting renewable energy, clean transportation, solar, hydro, and energy efficiency initiatives in developing countries.
- **Mexico's Green Bond:** Mexico has successfully issued its first green bond specifically for low-carbon transport, demonstrating the feasibility of this financing mechanism for developing economies with sufficient creditworthiness and robust financial markets.

The significant creditworthiness barrier for many LICs highlights a critical disconnect between global climate finance ambitions and the financial realities of vulnerable nations. While green bonds represent a promising market-based mechanism for financing climate-smart infrastructure, the observation that most LICs lack the creditworthiness to issue green bonds directly, relying instead on DFIs as intermediaries, points to a systemic challenge. This creditworthiness gap limits the direct flow of

private climate finance to the countries that often need it most for adaptation and mitigation efforts. It forces reliance on intermediation, which can add layers of cost and complexity, potentially slowing down the deployment of crucial climate-resilient infrastructure. To truly scale green bond financing for developmental infrastructure in LICs, the G-20 should explore mechanisms to enhance country-level creditworthiness. This could involve targeted technical assistance for fiscal management, and the provision of guarantees from Multilateral Development Banks (MDBs) to improve bond ratings. Additionally, innovative financial structures, such as pooled bonds that aggregate smaller projects from multiple countries or regional issuance platforms, could provide the necessary scale and credit enhancements to attract international investors. This would enable more direct market access for LICs, ensuring that climate finance reaches its intended beneficiaries efficiently and at the scale required to meet climate and development goals.

7. The G20 as Arena and Actor - Key Bloc Positions, Mandate Clarity, and the Structural Reform Imperative

The advanced industrial economies - the United States, the major European members (Germany, France, Italy, and the EU as a composite actor), the United Kingdom, Japan, South Korea, Canada, and Australia - constitute the original institutional core of the G20, the primary proponents of a rules-based liberal international economic order, and the states whose aggregate financial, monetary, and technological capacity gives the forum its operational weight. They are net energy and goods consumers, acutely exposed to imported inflation and supply chain disruption, and they possess the deepest fiscal and monetary buffers of any G20 bloc. Their combined GDP represents the plurality of the G20's collective economic output, their central banks and regulatory bodies set the de facto standards for global financial governance, and their technology sectors generate the majority of the

innovation that defines the frontier of industrial competition.

However, the characterization of the advanced economies as a coherent bloc capable of acting in concert - an assumption embedded in much multilateral diplomacy - is analytically misleading in 2026, because the transatlantic cleavage between the United States and the European members is among the most consequential and structurally significant dividing lines in the entire forum. The United States under its 2026 presidency has emphasized affordable and secure energy supply chains, deregulation, and a narrowing of the agenda back toward core economic questions, and has shown marked skepticism toward the broader development and climate framing favored by recent G20 presidencies. The posture of the Trump administration toward multilateralism has been consistently transactional - engaging with multilateral forums when they advance US strategic or commercial objectives and withdrawing from or undermining them when they do not. This has specific operational consequences for the G20: US resistance to climate-related language in communiqués, skepticism toward concessional development finance commitments, and a preference for bilateral or minilateral arrangements over broad multilateral frameworks have each been documented across the 2025–2026 cycle.

The European members, by contrast, having absorbed the 2022 Russian gas cutoff directly and invested enormous political and financial capital in the REPowerEU transition, place a substantially higher weight on energy diversification, the climate dimension of energy security, and the maintenance of multilateral governance frameworks as such - not merely as instruments of immediate interest but as structural goods whose erosion imposes long-run costs on the stability of the international system. The EU's completion of its Critical Raw Materials Act legislative framework, the conclusion of the April 2026 EU-US Critical Minerals MoU, and the European Commission's continued investment in strategic autonomy through the European Green Deal industrial framework all reflect a European strategic posture that prioritizes structural resilience over short-run cost minimization in ways that the

current US administration does not systematically share. Japan and South Korea occupy a position between these two poles: deeply exposed to energy supply disruption given their import dependence, committed to the liberal international economic order as the condition of their postwar prosperity, but also structurally dependent on the US security guarantee in ways that constrain their ability to diverge from US positions on the most sensitive issues. Australia's positioning in the critical minerals domain is uniquely consequential: as the world's largest lithium producer and a significant rare earth miner, Australia is simultaneously a consumer-nation aligned with the Western bloc's security concerns and a producer-nation with a fundamental interest in the price and investment stability of the minerals it exports - giving it a distinctive bridging role between the consumer and producer blocs.

The transatlantic divergence is most consequential on three specific issues before this committee: the framing of critical mineral supply chain resilience (the US preferring bilateral and minilateral arrangements under the Minerals Security Partnership and IRA framework, Europe preferring a broader multilateral architecture that includes producer nations without political alignment requirements); the treatment of development finance in the resilience agenda (the US under its current administration showing limited appetite for new multilateral development commitments, Europe committed to the EU-backed Global Gateway initiative as an alternative to BRI financing); and the role of climate framing in energy security discussions (the US seeking to separate energy security from climate, Europe insisting on their structural integration). Delegates representing or engaging with either the US or European blocs must understand that this divergence is not a temporary diplomatic friction but a structural difference in political economy and strategic orientation that will shape the achievable space of any G20 outcome.

8. The Consensus Trap

The G20 operates on a consensus basis - unlike the UN Security Council, it has no formal voting mechanism, and its communiqués and declarations reflect the lowest common denominator of positions that all twenty-one members (nineteen countries plus the EU and AU) can endorse. This design was a deliberate choice reflecting the founders' belief that legitimacy and effectiveness in international economic governance require voluntary buy-in from major economies, and that a non-binding consensus-based format better generates that buy-in than a formal legally binding treaty structure. The logic is compelling in a world where the geopolitical distance between member positions is manageable - where the space of achievable consensus contains enough substantive ground to enable meaningful coordination.

The problem, in 2026, is that this condition is not reliably met on the agenda items that matter most. The 2026 G20 operates under conditions in which the geopolitical distance between member positions on energy policy, sanctions architecture, supply chain governance, climate finance, and development finance may be too large for meaningful consensus to be achievable. The Council on Foreign Relations projected 2026 as likely to be "a disruptive year for the G20 - characterized by transactional diplomacy, recurrent tension, and minimalist outcomes." This is not merely an observation about the current political cycle - it reflects a structural dynamic in which the G20's membership has grown to include states with fundamentally incompatible interests and political systems, and in which the consensus requirement that was designed to ensure legitimacy functions instead as a veto system that prevents substantive action. The result is what might be termed "consensus inflation" - an escalation in the hedging, ambiguity, and qualification of communiqué language sufficient to neutralize any specific policy commitment that a significant bloc of members opposes. The G20 communiqué produced under the India presidency's managed navigation of the Ukraine language - noting members' "different

views" rather than condemning aggression - is the canonical example, but the same dynamic applies to energy subsidy reform, carbon pricing, and debt restructuring obligations across multiple presidencies.

The consensus trap has a specific and damaging distributional consequence: because the consensus requirement gives effective veto power to the member most opposed to any specific measure, it systematically biases outcomes toward the interests of the states with the highest stake in preventing change - the largest fossil fuel producers on climate measures, the largest mineral processors on supply chain governance, the largest creditors on debt restructuring. It is, structurally, a mechanism that protects incumbency and resists the redistributive measures that the resilience agenda requires.

9. The Agenda Problem

The G20 began as a finance ministers' forum in 1999 and was elevated to Heads of State level in 2008 with an explicit mandate for macroeconomic and financial stability coordination. Over the subsequent fifteen years, its agenda expanded dramatically - driven partly by the genuine systemic interconnection of macroeconomic stability with health, climate, digital governance, and food security, and partly by the political dynamics of rotating presidencies in which each host country uses its agenda-setting authority to incorporate domestic policy priorities. The forum now operates across Energy, Climate, Anti-Corruption, Development, Digital Economy, Disaster Risk Reduction, Education, Employment, Environment, Finance, Health, Agriculture, Tourism, Culture, and Trade working groups - a scope expansion that represents an enormous increase in the forum's nominal policy ambition and a significant decrease in its per-issue analytical depth and political weight.

The Carnegie Endowment for International Peace noted that the forum "will have to confront the ghosts of its legacy and past ambitions, the realities of its current limitations amid geopolitical

contestation, and the uncertain contours of its future role." The agenda inflation problem is not merely one of organizational efficiency - it is a strategic coherence problem. A forum that attempts to produce consensus-based outcomes on twenty-plus distinct policy domains simultaneously is a forum that produces minimalist outcomes on all of them, because the political capital available for building coalitions, making concessions, and resolving disagreements is finite, and it is being deployed across too many issues to be concentrated on the ones where concentration would be most consequential.

The analytical question - whether the G20 should focus narrowly on its original macroeconomic and financial mandate, recovering effectiveness and clarity at the cost of nominal scope, or maintain its broader ambition while accepting lower-density outcomes - is not merely an organizational question. It is a question about the kind of institutional architecture the world needs in a period of polycrisis: whether the appropriate response is a forum that attempts to integrate all dimensions simultaneously or a reformed macroeconomic coordination body that concentrates its comparative advantage in financial and economic governance while deferring to specialized multilateral bodies on the domains where those bodies have more appropriate mandates, deeper expertise, and more suitable membership.

10. The Provider-Beneficiary Gap: The Most Consequential Political Architecture Problem

The most analytically precise and politically under-addressed structural problem in the G20's current functioning is what may be termed the "provider-beneficiary gap" in resilience capacity. The states that bear the cost of maintaining global economic resilience - by holding strategic reserves, maintaining redundant supply capacity, absorbing the price premium of de-risked supply chains, or providing the financial buffers that stabilize the global financial system under stress - are not

systematically the same states that benefit from that resilience when it is invoked. Advanced G20 economies invest in strategic petroleum reserves, maintain overcapacity in critical supply chains, and provide the fiscal firepower behind IMF lending facilities - and in doing so, they generate positive externalities for the entire global economy, including the developing economies that contribute less to these buffers and benefit from them disproportionately.

This gap is not politically sustainable in the long run, because the willingness of premium-payers to continue absorbing the cost of resilience they do not fully internalize will erode under domestic political pressure - as the Trump administration's explicit "America First" framing of its G20 presidency makes vividly clear. The central political task of any durable G20 resilience framework is therefore to close this gap sufficiently that the arrangement is self-enforcing - to create mechanisms through which the principal beneficiaries of global economic stability contribute meaningfully to the cost of maintaining it, and through which the principal providers of that stability receive sufficient return to sustain their domestic political support for the enterprise. This is not a technical problem; it is the fundamental political economy problem of international economic governance, and the G20's failure to address it structurally is the most important reason why its resilience frameworks remain fragile and subject to periodic unraveling under domestic political pressure.

11. Structural Clarity as the Precondition for Everything Else

The analysis in this sub-agenda arrives at a conclusion that is both analytically straightforward and politically difficult: the G20's capacity to deliver on every other item of this committee's agenda - energy security architecture, emergency reserve coordination, transition-security compatibility, critical minerals governance, supply chain resilience, macroeconomic stabilization, and debt distress relief - depends ultimately on whether the forum itself is structurally capable of translating shared

analytical understanding into coordinated political commitment. A forum that is paralyzed by the consensus trap on its most consequential agenda items, diffused across twenty working group tracks rather than concentrated on the economic coordination that is its comparative advantage, and structured in ways that systematically protect incumbent interests against redistributive resilience measures, will not deliver the coordinated multilateral responses that the 2026 crisis demands, regardless of the analytical quality of its deliberations.

The committee's most consequential contribution will therefore be to identify not only the right policy responses to the specific agenda items but the right institutional conditions under which those policy responses can be produced - and to advocate, with analytical precision and political realism, for the structural reforms that would restore the G20's capacity to function as what it was designed to be: the premier forum for international economic cooperation at a moment when that cooperation is more urgently necessary than at any point in the forum's history.

12. Recommendations for an Effective G-20 Response

These are offered as the analytical spine for negotiation rather than as a fixed programme, and they follow directly from the analysis above. Delegates are expected to contest, refine, and extend them.

First, pre-commitment over discretion. Because cooperative will contracts when shocks arrive, the durable arrangements are those agreed and capitalised in advance, when the premium is cheap and the threat is abstract. Coordinated strategic-reserve release rules, agreed in advance and triggered automatically by defined conditions, convert a future political decision that the paradox predicts will not be taken into a present technical commitment that executes on its own. The negotiating question is not whether to cooperate in the next crisis but how much capacity to capitalise before it.

Second, transparency as the cheapest form of resilience. Much of the cost in each episode examined

flowed from informational failure rather than physical scarcity, since the repricing of dependence is driven by uncertainty about how fragile concentrated flows truly are. Shared, credible, and timely data on inventories, flows, and chokepoint conditions lowers the volatility that drives panic repricing, and information is the rare resilience instrument whose premium is low and whose provision does not trigger the security reclassification that physical redundancy invites. A G-20 monitoring and early-warning function for energy and supply-chain stress is among the most negotiable items available to this committee.

Third, distributional credibility as a precondition for systemic stability. Because fragmentation's costs and the burden of friend-shoring fall disproportionately on developing economies, any framework that protects only the resilience of the largest economies will accelerate the regionalisation it seeks to prevent. Extending buffer access, debt relief through a reformed Common Framework, and continued value-chain participation to the periphery is not charity within this analysis; it is the purchase of systemic stability, since each economy driven into a defensive bloc raises the fragility of the whole.

Fourth, restraint commitments with credible monitoring. The single most damaging category of response to a supply shock is the cascade of export bans and competitive hoarding that converts a localised shortage into a global one, as occurred with food exports in 2022. A G-20 commitment to refrain from export restrictions on critical goods during defined emergency conditions, paired with transparent monitoring of compliance, addresses the “beggar-thy-neighbour” dynamic at its source, though delegates must design it to be self-enforcing given the forum's lack of compulsory power.

13. Questions a Resolution Must Answer (QARMA)

1. How should the G-20 define the threshold of "regional instability" that triggers a coordinated economic response, and who determines that the threshold has been met?
 2. Through what mechanism can the forum pre-commit members to resilience measures, such as coordinated reserve releases or restraint on export bans, given that it produces no binding law?
 3. How should the cost of resilience, the redundant capacity, reserves, and higher-cost diversified supply, be distributed between the economies that can afford it and those most exposed to instability they did not cause?
 4. What role should the existing institutions, the International Monetary Fund, the International Energy Agency, the World Trade Organization, and the G-20's own Common Framework, play, and how should the gaps in their coverage be addressed?
 5. How should the committee treat critical-mineral and semiconductor concentration, which sit at the intersection of energy security, supply-chain resilience, and acute geopolitical risk?
 6. How can a coordinated framework remain credible and self-enforcing at the moment of crisis, when the cooperation paradox predicts that the will to comply will be weakest?
 7. How should the forum reconcile the divergent priorities of energy producers and consumers, of advanced and developing economies, and of the providers and beneficiaries of resilience capacity?
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14. CONCLUSION:

The agenda before this committee asks delegates to trace how a disturbance bounded in one region becomes a cost borne everywhere, and to determine what a forum without enforcement power can credibly do about it. The analysis offered here contends that energy security, supply-chain resilience, and macroeconomic stability are not three separate problems but three readings of a single condition: the cost the integrated economy incurs when its reliance on concentrated, efficiency-maximising dependence is exposed to instability. The European gas shock, the Red Sea disruption, and the slow fragmentation of the global economy differ in tempo and medium but share this mechanism, and in each the difference between a manageable repricing and a systemic crisis turned on the availability of substitution options and the buffers that happened to exist when the shock landed.

The deeper finding concerns cooperation itself, and it is the finding most likely to elude a superficial treatment of the agenda. The instinct that instability should call forth more cooperation is precisely inverted by the structure of the problem: instability raises the value of cooperation while destroying the conditions that sustain it, so that the moment of greatest need is the moment of least supply. This cooperation paradox transforms the committee's objective. The realistic aim is not to generate cooperation on demand during a crisis, which will fail, but to design arrangements that capitalise resilience in advance, that substitute shared information for absent trust, and that distribute protection widely enough to be self-enforcing. The G-20 has the convening weight of the systemically important economies and none of the coercive power of a state, and the test of this committee is not whether it produces language that every delegation can accept, but whether it produces a framework that matches the actual structure of the problem it confronts. Delegates who internalise that distinction will serve their committee well.